

In-Kind Transfers as Insurance[†]

By LUCIE GADENNE, SAMUEL NORRIS,
MONICA SINGHAL, AND SANDIP SUKHTANKAR*

Households in developing countries often face variation in the prices of consumption goods. We develop a model demonstrating that in-kind transfers will provide insurance benefits against price risk if the covariance between the marginal utility of income and price is positive. Using calorie shortfalls as a proxy for marginal utility, we find that this condition holds for low-income Indian households. Expansions in India's flagship in-kind food transfer program not only increase caloric intake but also reduce caloric sensitivity to prices. Our results contribute to ongoing debates about the optimal form of social protection programs. (JEL D12, H53, I18, I38, O12, O15)

A common feature of many developing countries is variation in the prices of staple commodities across space and over time. Price variation may leave households, particularly those with low incomes, vulnerable to risk in their ability to meet even basic consumption needs. In this paper, we consider the implications of price risk for the design of social protection programs, with a particular focus on the role of in-kind transfers. In-kind transfers are a critical component of social protection around the world: approximately 44 percent of safety net beneficiaries receive in-kind transfers (Honorati, Gentilini, and Yemtsov 2015), and over 90 percent of low-income countries have social protection programs that include in-kind transfers (World Bank 2014). We explore a novel benefit of such transfers by demonstrating both conceptually and empirically that when local commodity prices vary, in-kind transfers can deliver welfare gains to beneficiary households by providing insurance value in addition to pure redistribution. More broadly, our results highlight the importance of considering insurance in the design of social safety net policies.

Variation in local commodity prices in developing countries has been well documented. For example, prices can vary due to weather-related shocks to local

*Gadenne: Queen Mary University of London, Institute for Fiscal Studies, BREAD, and CEPR (email: l.gadenne@qmul.ac.uk); Norris: University of British Columbia (email: sam.norris@ubc.ca); Singhal: University of California, Davis and NBER (email: msinghal@ucdavis.edu); Sukhtankar: University of Virginia, JPAL, BREAD, and NBER (email: sandip.sukhtankar@virginia.edu). Rema Hanna was the coeditor for this article. We thank the coeditor, Mia Giuriato, Diti Jain, Lydia Kim, Jenny Le, Eric Robertson, Elizabeth Spink, and Everett Stamm for excellent research assistance, Vikas Dimble for help with obtaining data, and Oriana Bandiera, Amy Finkelstein, Peter Ganong, Maitreesh Ghatak, Seema Jayachandran, Remy Levin, Lee Lockwood, Karthik Muralidharan, Paul Niehaus, Ben Olken, Krishna Pendakur, Debraj Ray, Florian Scheuer, Sheetal Sekhri, and numerous seminar participants for comments. We are grateful to the International Growth Center (IGC), Harvard University's Asia Center, and the Bankard Fund for Political Economy for financial support.

[†]Go to <https://doi.org/10.1257/aer.20220822> to visit the article page for additional materials and author disclosure statement(s).

production, often exacerbated by a lack of market integration (Atkin 2013; Allen 2014).¹ The implications of this price risk for households and for optimal policy, however, are less clear. As prior work (e.g., Waugh 1944) has demonstrated, the welfare consequences of price variability are theoretically ambiguous. Intuitively, absent storage technology, households face a trade-off between the cost of smoothing consumption when prices are high and the gains from substitution toward cheaper consumption when prices are low. In addition, other factors such as income may covary with prices, affecting the welfare consequences of price volatility.

We begin with a simple model to demonstrate that when prices vary across states of the world, the optimal policy will provide price-indexed transfers that equalize marginal utility of income across states. The marginal utility of income and therefore optimal transfers may theoretically increase or decrease with respect to price. Households will prefer transfers that increase with respect to price as long as a simple condition holds: the covariance between the marginal utility of income and price is positive.

We next demonstrate that when this condition holds, in-kind transfers will provide positive insurance value to the beneficiary household in addition to the welfare gain from pure redistribution. This occurs because the value of in-kind transfers rises automatically with the local market price of the transferred good. These insurance benefits may be particularly relevant when there are logistical challenges—such as a lack of high-frequency local price data—that constrain the price indexing of cash transfers.²

Importantly, the *reason* that marginal utility of income might be higher in high-price states is not relevant for our test. It might be because of higher prices directly, or because incomes tend to be lower in high-price states of the world, or for some other reason. This means that even if a causal estimate of the effect of prices on marginal utility were available, it would not be appropriate for determining the optimal form of social protection program. Instead, our test would still rely on the covariance between prices and marginal utility, along the lines of a sufficient statistics approach (Chetty 2009) and analogous to recent work estimating the welfare effects of Medicaid (Finkelstein, Hendren, and Luttmer 2019). The correlation of interest is therefore unconditional: we do not seek to control for income shocks or other factors potentially correlated with prices. Indeed, since in some places incomes are likely to be higher in periods with high prices, it is easy to imagine settings where marginal utility is *lower* in high-price periods and in-kind transfers would be welfare-reducing relative to a constant-value transfer. A key advantage of our approach is that it does not require instruments for prices and so can be applied

¹In addition to these studies on India (Atkin 2013) and the Philippines (Allen 2014), a plethora of evidence exists on the lack of market integration and subsequent internal price variation in various other countries; for example, Uganda (Gollin and Rogerson 2014), Kenya (Burke, Bergquist, and Miguel 2019), Sierra Leone (Casaburi, Glennerster, and Suri 2013), and Peru (Sotelo 2020). One way to gauge the extent to which integration matters is provided by Atkin and Donaldson (2015), who show that the effect of distance on trade costs in Ethiopia and Nigeria is four to five times that in the United States.

²It is well-known that “community price surveys in developing economies are either absent or suffer quality problems” (Gibson and Rozelle 2005, p.69). In the Indian case, Khara (2014, p. 120) notes that “it is not ‘technically simple’ to index cash transfers; one needs to consider several factors—including local variation in prices, adequate infrastructure requirements to collect such information, and frequency of indexing the amount.”

in a wide variety of settings to understand when transfers should increase with local prices.

A challenge when implementing this test is to find an appropriate empirical proxy for the marginal utility of income. Our primary measure is an indicator for falling below minimum calorie requirements. The key assumption underlying this measure is that the marginal utility of income rises when households fall below minimum requirements. A vast literature has documented the negative consequences of calorie shortfalls, demonstrating long-run effects of even short-term episodes. Undernutrition has been shown to worsen health, human capital accumulation, and earnings.³ Calories have low substitutability across periods and with other types of (nonfood) consumption goods, so the curvature of utility with respect to calories is likely to be high, particularly for households close to minimum requirements.⁴

We implement the test in the context of India, using National Sample Survey (NSS) data from over 500,000 households across 28 states and 10 years. The average Indian household is exposed to substantial risk from food price fluctuations, as it spends 52 percent of its budget on food, with 9 percent spent just on rice—the most commonly consumed food staple and the focus of our analysis. We use an indicator for meeting minimum calorie requirements (MCR) from the Indian Council of Medical Research (ICMR) as well as calories per capita as (inverse) proxies for the marginal utility of income.⁵ Forty percent of households in our sample fall short of minimum recommended calorie intake guidelines.

Increases in the price of rice are significantly negatively associated with caloric intake: a 10 percent increase in the market price is associated with 1.3 percentage points fewer households (equivalent to 16 million individuals nationwide) meeting the MCR and a 0.7 percent decline in calories consumed by the average household. These findings are driven entirely by below-median-socioeconomic-status households. These results demonstrate empirically that high-price states are also high-marginal utility states for poorer households. In the context of the model, this positive covariance of marginal utility and price implies that one of the benefits of in-kind transfers is that they can provide positive insurance value, helping households smooth consumption in the face of price risk.

To what extent do in-kind transfer programs deliver such insurance benefits in practice, and how large would transfers need to be to smooth caloric sensitivity to prices fully? To answer these questions, we turn to an evaluation of India's flagship in-kind transfer program, the Public Distribution System (PDS). The PDS is one of the largest in-kind transfer programs in the world, providing food transfers to nearly a billion people in 180 million eligible households (Balani 2013). The program provides (mainly) rice and wheat every month at substantially below-market prices ("PDS prices") through a network of over 500,000 designated shops. However, the program has been criticized for corruption and mistargeting (Niehaus et al. 2013;

³ For a summary of the medical literature, see Victora et al. (2008); for literature in economics, see Currie and Almond (2011).

⁴ We lack the local price measures for most nonfood consumption needed to construct an accurate marginal utility measure based on total real consumption; see Section ID for further discussion.

⁵ We use MCR as shorthand for the ICMR's caloric guideline for the "sedentary" (lowest) level of exertion calculated by age and gender and aggregated to the household level.

Dreze and Khera 2015), and the limited rigorous evidence on its causal effects is mixed (Kochar 2005; Tarozzi 2005; Kaul 2018; Shrinivas et al. 2018).⁶

We examine the causal effects of the PDS on caloric intake and the calorie-price relationship using newly collected administrative data on state-level PDS policy changes between 2003 and 2012, a period between major national policy changes. The PDS became a more important part of the social safety net over this period (Khera 2011). Expansions in eligibility doubled the number of households receiving PDS grains, while decreases in PDS prices increased the real value of the transfer for each household. We use variation in the mandated PDS price as well as expansions in eligibility to instrument for *PDS value*: the quantity of rice obtained from the PDS multiplied by the difference between the market price of rice and the PDS price paid by beneficiaries.

We first document large effects of PDS expansions on the level of calories. A ₹100 increase in PDS value (the average nonzero PDS transfer) leads to a 10.7 percentage point increase in households meeting the MCR and a 6.4 percent increase in calories per capita.⁷ Overall, we estimate that PDS policy changes led to 40 million additional individuals meeting MCR thresholds over the study period.

We next examine the—previously unstudied—role of the PDS in reducing caloric sensitivity to local prices. A ₹100 increase in PDS value reduces the sensitivity of calories to market prices by 68 percent, with estimated sensitivity for the average household reaching 0 for a PDS transfer worth ₹147. This is only one-half larger than the average nonzero transfer, indicating that the PDS as implemented during our study period already provides a substantial amount of insurance against price risk. To the best of our knowledge, this is the first study examining the effect of transfer program receipt on the sensitivity of outcomes to prices. These results suggest a perhaps bigger role for the PDS in providing food security than previously understood and may be one reason why large numbers of beneficiaries report preferring in-kind food transfers from the PDS over equivalent value cash transfers (Muralidharan, Niehaus, and Sukhtankar 2017a; Das and Sethi forthcoming). Indeed, during the COVID-19 crisis, the PDS assumed an even more important role: not only as the main food security and social welfare program but explicitly as a bulwark against local food price shocks (Roy, Boss, and Pradhan 2020).

To alleviate concerns about policy endogeneity and omitted variables bias, we demonstrate that trends in calories prior to eligibility expansions are flat. Our results are also robust to controlling for political cycles and the generosity of the National Rural Employment Guarantee Scheme (India's other major welfare program) and are similar when we restrict the sample to states that are not major suppliers to the PDS.

Finally, we provide empirical evidence against alternatives to the insurance mechanism, demonstrating that income effects, flypaper effects, and general equilibrium price effects are orders of magnitude too small to explain the observed findings. Our results on the effect of in-kind transfers in reducing caloric sensitivity to prices

⁶See also Li (2021) for evidence of the effects of PDS expansions on home production and Shrinivas, Baylis, and Crost (2019) for evidence of small effects on labor supply.

⁷These results suggest that the time period of study might be important. Our paper; Kaul (2018); and Shrinivas et al. (2018) find positive effects on nutrition and study later expansions as compared to the older studies that find little or no effect.

thus complement our sufficient statistics approach by providing direct support for the insurance mechanism posited in the model and by quantifying the magnitude of these effects.

This paper contributes to several literatures. First, we speak to a longstanding literature on household exposure to price variability and its consequences. This literature has generally assessed the welfare effects of price risk relative to price stabilization (Waugh 1944; Massell 1969; Turnovsky, Shalit, and Schmitz 1980; Finkelshtain and Chalfant 1991; Bellemare, Barrett, and Just 2013).⁸ While stabilization and dual pricing policies are still used, many critics have argued that they are both expensive and ineffective (Rashid 2009; Bellemare, Barrett, and Just 2013). Moreover, the empirical literature on price risk is limited (Bellemare and Lee 2016), and to the best of our knowledge, previous studies have not considered the possibility of insuring against—rather than attempting to reduce—price variability. In addition, while the previous literature has taken a structural approach to estimate price risk preferences (see, e.g., Deaton 1989), we follow the sufficient statistics literature in deriving a test that requires the estimation of a simple empirical parameter. This allows us to determine whether households value insurance without directly parameterizing risk aversion or other components of the economic environment, such as price-income correlations, access to credit or storage, or home production.

Second, a related literature examines the specific issue of price shocks and food security.⁹ Numerous studies have examined the effect of food price shocks on nutrition, with mixed findings.¹⁰ Our study complements this literature by demonstrating the implications of this empirical relationship *without* requiring an instrument for prices, a major challenge in this literature. Since our test does not rely on exogenous variation in prices, it can be easily generalized to other empirical contexts.

Third, we propose a novel rationale for in-kind transfers. Previous studies have proposed other motivations for in-kind transfers: they can improve targeting (Nichols and Zeckhauser 1982; Besley and Coate 1991; Lieber and Lockwood 2019), the well-being of nontargeted households by reducing market prices of transferred goods (Coate, Johnson, and Zeckhauser 1994; Cunha, De Giorgi, and Jayachandran 2018), and the efficiency of imperfectly competitive food markets under some conditions (Coate 1989; Jiménez-Hernández and Seira 2021). Others have argued that in-kind transfers are beneficial if beneficiaries given cash will maximize the “wrong” utility function, either due to intrahousehold conflicts or simply a paternalistic view (Currie and Gahvari 2008; Cunha 2014; Batista, Silverman, and Yang 2015). However, although some in the policy community have highlighted the potential insurance benefits of in-kind transfers (Kotwal, Murugkar, and Ramaswami 2011; Dreze 2011), this rationale has been largely unstudied in the academic literature.

⁸One exception is Newbery (1989), who compares price stabilization to rations (in-kind transfers).

⁹Barrett (2002, pp. 2108–09) reviews the literature on food security in general, emphasizing the importance of risk as an important component of food security but noting that “most of the literature nevertheless fails to address issues of risk and uncertainty.” A large literature has considered how producer choices may be distorted by food price risk and poorly integrated markets (see, for example, Fafchamps 1992; Saha and Stroud 1994; Barrett 1996) and Fackler and Goodwin (2001) for a review of this literature.

¹⁰A number of papers show that positive food price shocks lead to worse nutrition (for example, Brinkman et al. 2010 and the various World Food Programme studies cited therein) and welfare losses (Attanasio et al. 2013). However, a significant number of careful analyses also find nonexistent or positive relationships (Jensen and Miller 2008; Behrman, Deolalikar, and Wolfe 1988).

The influential and comprehensive Currie and Gahvari (2008) review of cash versus in-kind transfers does not even mention it, and papers that empirically test the impact of different transfer modalities have not considered insurance as a potential mechanism (Hidrobo et al. 2014; Moffitt 1989; McIntosh and Zeitlin 2021; Siu, Sterck, and Rodgers 2021). One exception is Gadenne (2020), who models the PDS as a nonlinear commodity tax system in which two potential advantages (relative to a linear commodity tax) are redistribution and insurance. Understanding the insurance benefits of in-kind transfers is critical because of their global prevalence and importance.¹¹

Finally, our findings have implications for larger ongoing debates regarding the appropriate design of social protection programs. While in-kind transfers remain common around the world, numerous recent studies have highlighted the benefits of unconditional cash transfers (Haushofer and Shapiro 2017; Banerjee, Niehaus, and Suri 2019).¹² One important rationale for cash transfers is the textbook argument that beneficiaries themselves will be indifferent between cash and in-kind if transfers are inframarginal and will prefer cash if in-kind transfers are marginal and distort the consumption bundle. We contribute to this literature by demonstrating that inframarginal in-kind and cash transfers with the same expected value have different effects on household welfare when prices vary. Interestingly, and in contrast to the textbook intuition, beneficiaries themselves often report a preference for in-kind relative to cash in contexts as varied as Ecuador, India, Malawi, Kenya, and Ethiopia (Hidrobo et al. 2014; Khera 2014; Ghatak, Kumar, and Mitra 2016; Gentilini 2016; Shapiro 2019; Hirvonen and Hoddinott 2020). While there are many possible explanations for these stated preferences, beneficiaries who prefer food transfers to cash do frequently mention the fear of unstable prices as a reason for their preference (Khera 2014; Sabates-Wheeler and Devereux 2010).¹³ More generally, our findings highlight the important potential role of price risk for vulnerable households and the importance of designing social protection programs that mitigate exposure to this risk, through in-kind transfers or other policy tools, such as indexing cash transfers to prices.

The remainder of the paper proceeds as follows. Section I develops a framework for examining social protection programs in the context of price risk. Section II discusses the empirical setting and data. Section III presents evidence on price risk in India and provides an empirical test of the model. Section IV examines the effects of the PDS program on households and the extent to which it mitigates households' sensitivity to price risk. Section V concludes.

¹¹ Interestingly, Fetzter (2020) shows that NREGS (workfare) transfers eliminate the relationship between rainfall shocks and conflict in India. His idea that workfare can help households insure against adverse income shocks complements our more general argument regarding the insurance value of in-kind transfers.

¹² See also Blattman et al. (2017); Egger et al. (2019); and Ghatak and Muralidharan (2020).

¹³ There are of course many other reasons in-kind and cash transfers may differ from both the beneficiary and policymaker perspective. For example, stated household preferences may reflect differences between cash and in-kind in allocation mechanisms or intrahousehold dynamics. There may also be differences in administrative costs or corruption (Banerjee et al. 2021) or differences in whether the transfer affects not only the level but also diversity of food intake (see, e.g., Hidrobo et al. 2014; Ahmed et al. 2022; and Leroy et al. 2010).

I. Theoretical Framework

In this section, we present a simple theoretical model to examine the potential role of insurance in the presence of price variability. Our model builds on a well-established literature considering the welfare effects of price risk. Pioneering work by Waugh (1944); Massell (1969); Deschamps (1973); Hanoch (1977); Turnovsky, Shalit, and Schmitz (1980); and Newbery and Stiglitz (1981) studies theoretically the effect of commodity price risk on consumer welfare as well as the impact of price stabilization policies. More recent work generalizes their findings by considering households as both consumers and producers and extending to the multicommodity case (Finkelshtain and Chalfant 1991, 1997; Barrett 1996; Bellemare, Barrett, and Just 2013). Our contribution to this literature is twofold. First, we consider price risk in the context of the design of social protection programs, with a particular focus on in-kind transfers. Second, we derive an empirically testable statistic that can be used to determine the welfare effects of state-contingent and in-kind transfers without the need to specify the structural parameters of the model.

A. Optimal Insurance Policy

We begin with a simple model focusing on the welfare of a unitary household facing a varying price in one of several consumption goods. We derive three key results. First, the optimal insurance policy consists of price-indexed transfers that equate the marginal utility of income across states of the world. Second, optimal transfers may theoretically be increasing or decreasing with respect to price. Third, in-kind transfers will deliver positive insurance value if and only if the marginal utility of income is higher in the high-price states of the world.

We consider a household consuming several goods and assume that the price p_j of one of the goods, j , varies across states of the world with mean $\bar{p}_j$ and density $f(p_j)$. The prices of all other goods are fixed. For simplicity, we abstract from potential effects of transfers on market prices and assume that households treat inframarginal in-kind transfers as fungible.¹⁴ The household has income y and preferences characterized by the indirect utility function $v(\cdot)$. For expositional purposes, we assume that y is fixed but relax this assumption below.

We first characterize the optimal insurance policy: price-indexed (state-dependent) transfers. The optimal break-even menu specifies a set of transfers for each possible value of p_j , which we write $x(p_j)$, such that the expected value of these transfers, $\int x(p_j)f(p_j)dp_j$, is equal to zero. However, all the results derived below hold if the net expected value of the transfer is positive.

The optimal transfer $x(p_j)$ for a given price p_j is thus the one that maximizes $\int v(p, y + x(p_j))f(p_j)dp_j - \mu \int x(p_j)f(p_j)dp_j$, where μ is the marginal utility of income and p is the vector of all good prices. The first-order condition tells us that

¹⁴In-kind transfers may reduce market prices (Cunha, De Giorgi, and Jayachandran 2018; Jiménez-Hernández and Seira 2021) and cash transfers increase them (Filmer et al. 2021), though in practice, effects may be small (see Egger et al. 2019). Behavioral households could exhibit flypaper effects in spending from in-kind transfers (Hastings and Shapiro 2018). However, we find that both effects are negligible in our empirical context (see Section IVF).

the optimal menu equates the marginal utility of income $v_y(p, y + x(p_j))$ in all states of the world:

$$(1) \quad v_y(p, y + x(p_j)) = \mu, \forall p_j.$$

The optimal policy will transfer larger amounts to households in states with higher marginal utility of income. Optimal transfers $x(p_j)$ will therefore be increasing in the price if the marginal utility of income is itself increasing in the price. Taking the derivative of Roy's identity with respect to income, we can write the derivative of the marginal utility of income with respect to price in the following way:

$$(2) \quad v_{yp}(p, y + x(p_j)) = \frac{v_y(p, y + x(p_j))}{p_j} \alpha_j (\gamma - \eta_j),$$

where α_j is the budget share the household spends on good j , γ is the coefficient of relative risk aversion, and η_j is the income elasticity of demand for good j . The sign of this expression will depend on $(\gamma - \eta_j)$. Intuitively, if households are not very risk averse, they prefer transfers in the low-price state to take advantage of higher purchasing power. As risk aversion increases, the value of consumption smoothing increases, leading households to prefer transfers in the high-price state.¹⁵ This result parallels Turnovsky, Shalit, and Schmitz (1980), who show that households will be better-off with varying prices than with price stabilization if their demand elasticities are high relative to their risk aversion. The amounts transferred across states of the world are increasing in α_j : the higher the budget share spent on the good, the greater the sensitivity of marginal utility to price.

B. Extending the Model

A key advantage of this approach is that we do not need to explicitly specify all potential components of the utility function: because agents are optimizing, the derivative of marginal utility with respect to price will continue to be sufficient to assess the welfare effects of transfers.

As an illustrative example, we consider the case in which income covaries with local prices. This is likely since local prices themselves will be affected by local supply and demand conditions if there is a lack of market integration. Allowing household income to covary with prices, we obtain the following expression for the derivative of the marginal utility of income with respect to price:

$$(3) \quad v_{yp}(p, y + x(p_j)) = \frac{v_y(p, y + x(p_j))}{p_j} \left(\alpha_j (\gamma - \eta_j) - \gamma \frac{\partial y}{\partial p_j} \frac{p_j}{y} \right).$$

The additional term on the right-hand side captures the effect of allowing income to be correlated with prices: a positive derivative implies that high-price states

¹⁵ The higher the income elasticity η_j , the more consumption of the good is increasing with income, making income in the low-price states of the world relatively more attractive. η_j also captures the possibility of substitution to other goods.

of the world are also high-income states of the world. If this term is positive and sufficiently large, the marginal utility of income will decrease with the price even if $\gamma > \eta_j$. Intuitively, if price and income are positively correlated, a dollar in the high-price state generates less marginal utility than in the baseline framework. This formulation allows an arbitrary correlation between income and prices, which we might expect to be different (for example) between households who are producers versus consumers of the good.

However, the form of optimal transfers continues to be determined solely by the derivative of the marginal utility of income with respect to price. Specifically, transfers will be increasing with respect to price if and only if this derivative is positive.

$$(4) \quad x'(p_j) > 0 \Leftrightarrow v_{yp}(p, y + x(p_j)) > 0.$$

v_{yp} reflects risk aversion and income elasticity and allows for price-income correlations. This derivative will also capture the effects of potential correlations between p_j and the prices of other goods,¹⁶ as well as other dimensions of household behavior such as savings, storage, and home production.

C. In-Kind Transfers

We now examine the role of in-kind transfers, which are both a key component of social protection programs in practice and may be of particular importance when governments face challenges in implementing the optimal state-contingent policy due to logistical constraints in the measurement of local prices.¹⁷ To determine the insurance value of in-kind transfers, we compare them to a fixed benchmark transfer that provides pure redistribution. We model the in-kind transfer as providing a fixed amount z of the good to the household, while the benchmark transfer provides the same amount $z\bar{p}_j$ to the household across all states of the world. Both transfers thus provide the same transfer to the household in expectation. For simplicity, we assume the in-kind transfer is inframarginal (the household consumes more than z of the good for all possible prices p_j).¹⁸ Finally, we assume that prices are not affected by the form of the transfer. We use the term “welfare” to refer to the expected utility of the household. We acknowledge, however, that a social planner might also want to consider the social cost of provision, which could differ based on the form of the transfer.¹⁹

¹⁶Until now, the j indexing on the derivative of marginal utility with respect to price has been implicit since only the price of the in-kind good has varied; to account for other prices varying as well, one would index the derivative v_{yp_j} .

¹⁷Previous work has argued, for example, that observing local prices in real time and at high frequency is often infeasible in many developing country contexts (Gibson and Rozelle 2005; Khera 2014).

¹⁸This assumption holds for 93 percent of households in our empirical context. Our results below will also hold if transfers are marginal but households can engage in resale at the market price. Otherwise, the total welfare gain from in-kind transfers will be reduced as a result of distortion to the consumption bundle.

¹⁹Estimating the social cost is challenging: in-kind procurement often interacts with distortionary production subsidies and purchase guarantee schemes. In some cases, transfers may be financed externally through aid organizations. In addition, in-kind transfers may be administratively costly to implement or prone to corruption (Banerjee et al. 2021).

The welfare effect of the benchmark transfer can then be written as

$$(5) \quad W_B = z \bar{p}_j \int v_y(p, y) f(p_j) dp_j,$$

and the welfare impact of the in-kind transfer as

$$(6) \quad W_K = z \bar{p}_j \int v_y(p, y) f(p_j) dp_j + z \int v_y(p, y) (p_j - \bar{p}_j) f(p_j) dp_j.$$

Plugging (5) into (6), we obtain

$$(7) \quad W_K - W_B = z \int v_y(p, y) (p_j - \bar{p}_j) f(p_j) dp_j,$$

where the term on the right-hand side is simply the transfer amount z multiplied by the covariance between the marginal utility of income and prices. Using a linear approximation of $v_y(p, y)$ with respect to prices around the vector of mean prices $\bar{p}$, we obtain

$$(8) \quad W_K - W_B \approx z v_{yp}(\bar{p}, y) \int (p_j - \bar{p}_j)^2 f(p_j) dp_j.$$

Expression (7) shows that in the presence of price risk the in-kind transfer is not equivalent to the benchmark transfer from the household perspective, even though the expected monetary value of both transfers is the same. Moreover, as long as the covariance between the marginal utility of income and prices is positive (or, equivalently, as long as the derivative of the marginal utility of income with respect to price is positive—see expression (8)), the in-kind transfer is welfare improving with respect to the benchmark. Therefore,

$$(9) \quad W_K > W_B \Leftrightarrow v_{yp}(p, y) > 0.$$

This is because when the covariance is positive, the household's optimal insurance menu is one in which transfers are increasing with respect to price. The in-kind transfer provides this insurance function by delivering more purchasing power in the high-price states, thereby better approximating the optimal policy than a state-invariant transfer, which provides pure redistribution. In online Appendix A1, we demonstrate that the in-kind transfer will be equivalent to the optimal transfer for particular parameter values but in general will underperform the optimal transfer because it scales the transfer value with respect to price only as a function of the in-kind transfer quantity, rather than as a function of the household's preferences.

D. Model Implementation

In practice, we do not directly observe marginal utility of income and therefore rely on consumption-based measures as empirical proxies. Our main measure is an indicator for households failing to meet minimum calorie requirements. This allows us to capture total real food consumption—a substantial share of total consumption—in a single measure derived solely from quantity data. The identifying assumption is that an increased likelihood of failing to meet minimum requirements is associated with an increase in the marginal utility of income. This assumption is

likely to be satisfied since calories have low substitutability (both intertemporally and with nonfood consumption), and the curvature of the utility function with respect to calories near the threshold is likely to be high.

Note that we are unable to construct an accurate measure of total real consumption because we do not have local price measures for 25 percent of food consumption and 87 percent of nonfood consumption. However, if we deflate expenditure by the limited price vector available and use this as an outcome, we find very similar results to the calorie results presented below (see online Appendix A2.3 for a discussion of the data limitations and results).

An important interpretational consideration arises if the observed calorie gradient with respect to price reflects costly consumption smoothing mechanisms on the part of households (Chetty and Looney 2006). In this case, observing a positive relationship between the likelihood of failing to meet caloric thresholds and price is a sufficient but not necessary condition for in-kind transfers to improve household welfare through an insurance channel: observing no relationship could still be consistent with in-kind transfers allowing risk-averse households to be less reliant on costly smoothing behaviors.

II. Context and Data

A. Context

We examine the predictions of the model in the context of India, which is ideal for studying these issues for a number of reasons. First, as much prior research has documented, markets are not well integrated, and local prices are subject to volatility arising from (for example) weather shocks (Rosenzweig and Udry 2014). Substantial price differences persist across regions, and temporary shocks to local prices are frequent (Atkin 2013). Second, as we discuss below, a substantial share of the population fails to meet basic caloric requirements. Finally, India has one of the world's largest in-kind transfer programs: the Public Distribution System.

The PDS is one of India's oldest and most important antipoverty programs, dating back to several months before independence in 1947. The PDS provides goods such as rice at significantly below-market rates to eligible households via a widespread network of fair price shops.²⁰ The program operates much like in-kind transfer programs across the rest of the world: the government procures goods directly from producers in a few agricultural states and then sells them to households at below-market prices.²¹ The effective transfer is substantial: for example, the PDS price for rice was 35 percent of the market price on average over our study period and 10 percent today.²² Eligible households can buy up to a certain quantity of grains each month

²⁰The PDS also provides wheat, kerosene for cooking fuel, and less commonly sugar, salt, and other local grains.

²¹One explicit goal of the PDS is to provide a price floor for farmers selling agricultural products. Before the spring and winter harvests, the Commission for Agricultural Costs and Prices sets a guaranteed minimum price for key crops at which it will purchase from farmers if necessary. Geographic centralization of production—in 2016–2017, 78 percent of all rice procured was from the top 6 (out of 29) states (FCI 2018)—means that effects of the PDS on producers are concentrated away from most of our sample. In Table 8 we show that our results are similar when we exclude PDS-producing states.

²²Accounting for a below-market price the household must pay for the in-kind good modifies the implicit transfer amount—from $z\bar{p}_j$ in our baseline model to $z(\bar{p}_j - p_j^{PDS})$ —but does not affect the model test. In practice, the

based on entitlements listed on ration cards, although in practice, the PDS shops may not always have enough for all households to access their allocations.

Before 2000, eligibility was largely restricted to poor households, in particular those considered to be below poverty line (BPL). The PDS has grown more generous over the last 20 years, with large nationwide expansions in 2000 and 2013. In 2000, 6 million households became newly eligible, and PDS generosity was increased for the very poorest households. In 2013, the National Food Security Act further expanded eligibility to 75 percent of the rural population. Between these two federal changes, many states expanded their own PDS generosity.

B. Data

Our main source of data is the 59 through 68 rounds of the National Sample Survey, covering January 2003 through June 2012 (National Sample Survey Office of India 2005, 2006, 2007, 2008, 2009, 2010, 2012, 2014). This covers most of the period between 2000 and 2013, when the basic structure of the program stayed the same but generosity was dramatically increased in many states. We begin our sample period in 2003 because the NSS does not consistently identify many districts before the fifty-ninth round. June 2012 is the end of the survey period for the sixty-eighth round.

The NSS is a repeated cross-sectional survey that asks households about their expenditure in each of about 350 categories over the past 30 days. For a subset of these categories where the units are well defined, it also records the quantity consumed. In addition, the survey contains basic demographic information like household size and composition, religion, caste, landholding, assets, education, and occupation. We categorize households by the year-quarter in which they were surveyed.

As is standard for empirical work in India, we exclude Union Territories and Delhi due to small sample sizes in these regions (see, for example, Imbert and Papp 2015). The sixty-fifth and sixty-seventh rounds did not include the expenditure survey, so we do not observe household outcomes in the periods July 2008 to June 2009 and July 2010 to June 2011. In total, our sample includes 524,911 households spread across 28 states.

We use the NSS in two main ways. First, we follow Deaton and Tarozzi (2005) and use unit values—expenditures divided by quantities—as the basis to measure local rice prices. Second, we use the NSS to construct measures of caloric intake, which we use as outcome variables. Online Appendix A2 provides further details on the NSS and data construction.

Unit Values.—India lacks measures of prices that are for individual items, cover the entire country, and vary at the local level. To overcome this challenge, we construct unit values from expenditure and quantity information:

PDS prices are so low that the system is close to free distribution (see Figure 2, panel A); it is extremely rare for households to not be able to afford “co-pays.” Rather, the main reasons for not getting PDS grains are related to technical/bureaucratic issues, and conditional on getting grain, beneficiaries obtain 94 percent of their entitlements (Muralidharan, Niehaus, and Sukhtankar 2023).

$UV_{ijt} = \frac{\text{expenditure}_{ijt}}{q_{ijt}}$ for good j consumed by household i in time t .²³ Using unit values rather than prices is standard practice in the literature that uses the NSS (Subramanian and Deaton 1996; Deaton and Tarozzi 2005) as well as in work on rural prices elsewhere (Sotelo 2020). We remove observations that appear to result from transcription or data errors; see online Appendix A2.2 for more details.

The smallest consistently identified geographic units in the NSS are districts interacted with a rural/urban (“sector”) indicator, and most of our analysis conditions on fixed effects at this level. However, for sample size reasons, we measure local prices at a slightly higher level. Instead of districts, we use NSS regions, groupings of contiguous districts having similar geographical features, rural population densities and crop-pattern that are likely to face similar price shocks. The tenth percentile region-sector-quarter has 23 consumers of market rice, and the twenty-fifth percentile has 42, compared to 4 and 6 at the district-sector-quarter level (online Appendix Table A2). There are 140 region-sectors, and we measure prices using the mean unit value at the region-sector-quarter level.

A natural measure of prices is place- and time-specific average unit values. However, this approach is susceptible to bias arising from measurement error in either expenditure or quantities. In particular, classical measurement error in quantities would affect the left-hand side (caloric intake) and the right-hand side (average unit value) in opposite directions, resulting in negative bias. Alternatively, household income shocks that increase caloric consumption might also increase measured prices due to quality upgrading. This would cause positive bias.

To avoid a mechanical correlation between unobservable income or taste shocks that affect both outcomes and prices, we instead use a leave-out measure of prices:

$$(10) \quad p_{-i,jt} = \frac{1}{N_{r(i)} - 1} \sum_{k \in R(i) \setminus i} UV_{kjt},$$

where $R(i)$ is the collection of households in the region-sector in which household i is located and $N_{r(i)}$ is the total number of households in i 's region-sector. The correlation between this leave-out price measure and the simple average $p_{ijt} = \frac{1}{N_{r(i)}} \sum_{k \in R(i)} UV_{kjt}$ is extremely high; the coefficient on $p_{-i,jt}$ in a regression of p_{ijt} on leave-out prices and our baseline set of controls is 0.994. As a result, there is nearly no difference in the results if we use the non-leave-out price measures.²⁴

Although our price measures are by construction uncorrelated with individual-level demand shocks, one possible remaining concern is that unobserved region-sector-level taste or income shocks affect both demand and quality. In the most likely scenario, unobserved income shocks cause higher caloric intake and substitution into higher-quality rice, leading to upward-biased coefficients. To assess whether unit value prices accurately capture changes in the prices faced by

²³The NSS data separately record information for goods purchased from the market and the PDS, so we observe unit values separately for market goods and for goods purchased from PDS shops.

²⁴An alternative would be to instrument for p_{ijt} with $p_{-i,jt}$. However, given the first stage of 0.994, the results are nearly indistinguishable.

households, we therefore turn to the Rural Price Survey (RPS) (National Sample Survey Office of India 2013). The RPS is a survey of vendors at local markets and covers many of the same goods as the NSS. Unfortunately, however, it covers only a limited subsample of rural areas, and a lack of documentation makes it impossible to determine the sampling frame. We therefore do not use it for our main analysis. Nonetheless, within the overlapping sample (about 25 percent of the NSS sample areas), we find a correlation over time in NSS unit values and RPS prices for rice of nearly 0.60 (see online Appendix Table A4). To more directly assess the robustness of our results, we also show below that our results using RPS prices are identical to those using NSS unit values in the overlapping sample.

Household Characteristics.—Since our object of interest is the price risk faced by individual households over time, we control for permanent household characteristics (indeed, if the same households appeared in the NSS in different rounds, we would control for household fixed effects). The most important of these is household permanent income, which we proxy for using a socioeconomic status (SES) index. We construct this index by regressing log per capita expenditure on scheduled caste/scheduled tribe (SC/ST) status; fuel used at home; home ownership; occupation and education of household head; land possessed; the number of household members in the 18 bins defined by the intersection of age (0–17, 18–54, 55+), gender and education (below primary, primary, above primary); and district-sector-season; round; and period fixed effects. The SES index is the predicted value from this regression, standardized to have a mean of zero and a standard deviation of one. When we split the sample by above- and below-median SES, we construct the cutoff using NSS weights within state-rounds.

We use several other household characteristics as controls and as dimensions of heterogeneity to examine. We capture economies of scale in consumption using log household size. Religion and scheduled caste/scheduled tribe status (constitutional status for groups historically discriminated against) as well as the type of cooking fuel used all determine the type of food that households eat and therefore calories consumed. We use landholding as a proxy for households' ability to produce food commodities. We define landholding households as owning more than 0.01 hectares of land, which allows us to proxy for the ability to engage in agricultural production.

Calorie Requirements.—As discussed above, the relevant parameter for determining the optimal form of transfers is the correlation between marginal utility of income and prices. If this relationship is positive, (i) households will prefer transfers that increase with respect to price, and (ii) in-kind transfers will therefore deliver welfare gains to households relative to a fixed equal expected value cash transfer by providing insurance benefits. Our main empirical proxy for marginal utility of income is an indicator for whether the household fails to meet a minimum recommended caloric intake. We interpret an increased likelihood of failing to meet basic calorie requirements as associated with an increase in the marginal utility of income. We also examine calories per capita as an additional outcome.²⁵

²⁵ As discussed above, we cannot construct an accurate measure of real total consumption because the price vector for nonfood consumption is highly incomplete. However, the consumption results using the limited available price information are very similar to our calorie results; see online Appendix A2.3.

We estimate household-level caloric intake using the information on total consumption of each item (including consumption from the market, the PDS, and home production) combined with estimates of the caloric value of each item.²⁶ To contextualize caloric consumption, we rely on age $\times$ gender specific guidelines for caloric intake from the Indian Council of Medical Research (Rao and Sivakumar 2010) and calculate the total household requirement. The ICMR provides separate caloric guidelines for different levels of exertion: sedentary, moderate work, and heavy work. We focus on the lowest of these, the “sedentary” guideline, to define the minimum calorie requirement by age and gender.²⁷ On average, individuals consume 2,097 calories per day, while the ICMR estimates that 1,904 would be necessary on average given our NSS sample’s age-gender composition. On average, only 61 percent of households meet the MCR (69 percent of above-median-SES households and 56 percent of below-median; see Table 1).²⁸

III. Empirical Test of Optimal Form of Transfers

A. Price Exposure and Variability

Indian households face considerable potential exposure to price risk because budget shares of staple goods are large. Table 1 shows that the average household spends 52 percent of its budget on food and 9 percent on rice alone. In our empirical analysis, we focus on rice because it comprises a substantial portion of household food expenditure, it is consumed throughout the country, and it is one of the main goods provided through the PDS system. In line with our assumption of inframarginality in Section I, only 6.6 percent of all households and 8.8 percent of below-median-SES households consume rice from the PDS but not from private sources during the 30-day recall period.²⁹

We next examine variation in market rice prices over time and across areas (Table 2). Deflating by the all-India CPI, the mean price of rice is ₹9.73 per kilogram.³⁰ Taking out district-sector-season fixed effects, the standard deviation of the residual is 1.17. We then include year-quarter fixed effects to capture common shocks across areas.³¹ The residual standard deviation decreases to 1.03. Household characteristics do not explain any of the remaining variation: the standard deviation

²⁶These estimates are included by the NSS in the 55-round data release (NSSO 2001), which adopted them from Gopalan, Sastri, and Balasubramanian (1989).

²⁷Eli and Li (2020) show that caloric requirements vary across seasons in our context. Using the lowest recommended caloric intake defined by the ICMR for “sedentary” exertion ensures that decreases in calories around that threshold are associated with lower utility regardless of the actual type of work undertaken by household members.

²⁸We do not have data on consumption by individual and hence are restricted to calculating calories at the household level (and reporting results per capita for convenience; we also show results per adult equivalent in online Appendix Tables A5 and A13, which are similar to the per capita results). Of course, calories may be unevenly distributed within households, implying that individuals may not meet MCRs even in households that consume sufficient calories overall (Brown, Calvi, and Penglase 2018; D’Souza and Tandon 2019).

²⁹This is true throughout our period: even after the expansion of PDS generosity beginning in 2008, 89.2 percent of below-median SES households were inframarginal.

³⁰We convert all nominal values to 1999 rupees using the all-India CPI from the World Bank. US\$1 was ₹43 in 1999.

³¹We also control for NSS round effects to account for any potential differences in survey procedure or instruments. Because not all households are surveyed within the scheduled time, NSS round fixed effects can be included separately from year-quarter fixed effects.

TABLE 1—SUMMARY STATISTICS: DAILY CALORIC CONSUMPTION

	Food share of expenditure (1)	Rice share of expenditure (2)	Total calories per capita (3)	Per capita MCR (4)	Met MCR (5)
Overall	0.52 (0.13)	0.09 (0.09)	2,097 (633)	1,904 (231)	0.61 (0.49)
Below-median SES	0.55 (0.11)	0.10 (0.10)	1,976 (548)	1,861 (226)	0.56 (0.50)
Above-median SES	0.47 (0.13)	0.06 (0.06)	2,295 (707)	1,974 (222)	0.69 (0.46)
Rural	0.54 (0.12)	0.10 (0.10)	2,097 (633)	1,886 (228)	0.62 (0.49)
Urban	0.45 (0.13)	0.06 (0.06)	2,097 (633)	1,952 (232)	0.57 (0.49)
Rural landless	0.54 (0.12)	0.09 (0.09)	2,002 (636)	1,877 (245)	0.55 (0.50)
Rural landowning	0.54 (0.11)	0.10 (0.10)	2,135 (627)	1,890 (221)	0.65 (0.48)

Notes: The table shows summary statistics for daily household calorie consumption. Column 1 reports summary statistics for share of household-reported expenditure on all combined food items. Column 2 reports summary statistics for share of household-reported expenditure on market rice. Column 3 reports summary statistics for household calories per capita, estimated from the quantity and average caloric content of all food items consumed by the household during the survey recall period. The upper and lower 0.1 percent of calories per capita are trimmed to adjust for implausibly extreme calorie figures. Column 4 reports summary statistics for household average minimum calorie requirement, which is calculated as the average MCR of all household members based on the household demographic composition and recommended caloric intake guidelines published in 2012 by the Indian Council of Medical Research. Column 5 reports summary statistics for an indicator that the caloric consumption of the household met or exceeded its average MCR. Standard deviations in parentheses.

TABLE 2—SUMMARY STATISTICS FOR MARKET RICE PRICES

	Mean	SD			
	(1)	(2)	(3)	(4)	(5)
Overall	9.73	2.25	1.17	1.03	1.03
Below-median SES	9.37	2.07	1.14	1.02	1.02
Above-median SES	10.31	2.40	1.19	1.01	1.01
Rural	9.37	2.12	1.14	1.02	1.02
Urban	10.68	2.31	1.25	1.05	1.05
Rural landless	9.59	2.02	1.15	1.01	1.01
Rural landowning	9.28	2.15	1.11	1.00	1.00
District-sector-season fixed effects		No	Yes	Yes	Yes
Period fixed effects		No	No	Yes	Yes
Controls		No	No	No	Yes

Notes: This table shows mean unit values for rice from NSS survey data 2003–2012. Unit values of rice are the means of deflated average rice expenditure per kilogram across all households from the same region-sector-quarter. In reporting subgroup prices we use the same overall region-sector-quarter mean; the differences across these rows therefore reflect differences in the places and times where different groups reside. All unit values are measured in 1999 rupees. Controls include log household size, SC/ST, land ownership, religion, cooking fuel, and socioeconomic status index. All households owning 0.01 hectares of land or greater are classified as landowning. Household-level SES is the predicted value from a regression of log expenditure per capita on permanent household characteristics, with district-sector-season and period fixed effects. Period fixed effects include year-quarter and NSS round fixed effects.

is unchanged when we include household controls and the SES index. In our main specifications, we use the residual variation in the final column to estimate caloric responses to price variability to focus on the component of variation that is idiosyncratic across time and space. In practice, this provides a conservative estimate of the true price risk faced by households since they may not actually be able to smooth common cross-area or seasonal shocks, and we provide specifications with varying sets of fixed effects below.

The remaining rows of Table 2 show the same summary statistics by demographic groups. Unsurprisingly, the average prices faced by urban households are higher than rural households, as are average prices for above-median-SES households compared with below-median.

B. Empirical Specification and Results

Our primary outcome measure is an indicator for the household falling below the MCR; we also examine calories per capita as an outcome. Ex ante, it is not obvious that high-rice-price states will be associated with lower caloric intake: high-price states could also be high-income states, and households may be able to substitute toward other goods. In addition, the relationship is estimated given existing antipoverty programs and household smoothing mechanisms, such as self-insurance (for example, drawing down savings) and informal insurance. Finding a nonzero relationship implies that these smoothing mechanisms cannot fully insure price risk, leaving a role for state-contingent transfers. The sign of this relationship—which in turn will determine whether households prefer transfers that increase with respect to price and consequently, whether in-kind transfers deliver insurance benefits—is therefore an empirical question.

In Table 3 we regress the calorie outcome c_{it} on log leave-out market rice prices $p_{-i,rt}$:

$$(11) \quad c_{it} = \beta p_{-i,rt} + X_{it}\lambda + \delta_{da} + \tau_t + \phi_{round} + \varepsilon_{it},$$

where i indexes household, d indexes district-sector, r indexes region-sector, a indexes agricultural season (quarter of year), and t indexes the year-quarter in which the survey took place. We control for district-sector $\times$ season fixed effects (δ_{da}) to account for place-specific agricultural cycles, year-quarter fixed effects (τ_t) for national changes in policy and economic growth, and NSS round fixed effects (ϕ_{round}). We additionally control for household characteristics X_{it} including log household size, religion and scheduled caste/scheduled tribe status, land ownership, cooking fuel used, and the SES index. Regressions are estimated using NSS weights, and standard errors are clustered at the region-sector level, the level of our price variation.

We want our estimates to capture the empirical relationship between market rice prices and the marginal utility of income, allowing covariance of income and prices as well as substitution across goods in response to changes in relative prices. We therefore do *not* control for current household expenditure or other commodity prices. These estimates will capture the average correlation between prices and our

TABLE 3—MEETING THE MINIMUM CALORIE REQUIREMENT AND MARKET PRICES

	All districts				RPS districts	
	(1)	(2)	(3)	(4)	(5)	(6)
log market rice price	−0.131 (0.041)	−0.112 (0.044)	−0.131 (0.041)	−0.165 (0.042)	−0.301 (0.078)	−0.304 (0.081)
District-sector fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
District-sector-season fixed effects	Yes	Yes	No	Yes	Yes	Yes
Period fixed effects	Yes	Yes	Yes	No	Yes	Yes
HH controls	Yes	No	Yes	Yes	Yes	Yes
SES controls	Yes	No	Yes	Yes	Yes	Yes
Observations	524,911	524,911	524,911	524,911	175,065	175,065

Notes: This table displays regressions of an indicator for meeting minimum calorie requirement on log leave-out unit value market rice prices from NSS survey data 2003–2012. Column 6 measures prices using the Rural Price Survey; all other columns use mean NSS unit values. Columns 5 and 6 are restricted to districts in which RPS data are available. Household controls are log household size, SC/ST, land ownership, religion, and cooking fuel. Household-level SES is the predicted value from a regression of log expenditure per capita on permanent household characteristics, with district-sector-season and period fixed effects. Period fixed effects include year-quarter and NSS round fixed effects. Standard errors in parentheses and clustered at the region-sector level.

proxy for the marginal utility of income given any existing household insurance mechanisms as well as access to social safety nets, including the PDS.

Column 1 of Table 3 shows our preferred specification, regressing the likelihood of meeting the MCR on log market rice prices, controlling for district-sector-season fixed effects, year-quarter and NSS round fixed effects, the SES index, and household controls. A 10 percent increase in the price of rice is associated with a decrease in the likelihood that households meet the MCR by 1.31 percentage points, and this effect is significant at the 1 percent level. The SES index and household controls are meant to capture household permanent income and characteristics that are likely to affect diet and calories directly. However, if we exclude these, we still see a decrease in the likelihood of meeting MCR of 1.12 percentage points for every 10 percent increase in the rice price (column 2). We lose some precision, but the estimates are still significant at the 5 percent level. In column 3, we include district-sector fixed effects but not district-sector $\times$ season fixed effects to allow for seasonal variation in our price measure. The coefficient is identical to our baseline estimate, indicating that caloric shortfalls have similar sensitivity to seasonal and nonseasonal sources of price variation. In column 4, we remove year-quarter and NSS round fixed effects. The coefficient increases in magnitude (though the difference is not statistically significant), suggesting that households are not able to easily insure against shocks that are common across areas.

Finally, we compare our main estimates to results using prices from the Rural Price Survey. Again, we do not use the RPS for our main analysis because it is only conducted in a selected subsample of rural areas and because documentation on the sampling methodology is incomplete. However, it has two important advantages in addressing concerns both about measurement error in prices and potential substitution across rice quality varieties. First, prices are measured in local markets by surveyors and are not derived from household expenditure or quantity data. Second, the RPS is used to construct consumer price indexes and is therefore meant to capture

TABLE 4—MEETING THE MINIMUM CALORIE REQUIREMENT AND MARKET PRICES BY SUBSAMPLE

	By median SES		By census region		Rural by landowning	
	Below (1)	Above (2)	Rural (3)	Urban (4)	Landless (5)	Landowning (6)
log market rice price	−0.234 (0.056)	−0.035 (0.039)	−0.191 (0.053)	−0.024 (0.058)	−0.302 (0.090)	−0.157 (0.049)
Equality of effect (<i>p</i> -value)		0.00		0.03		0.09
Observations	211,796	313,115	316,234	208,677	63,614	252,620

Notes: This table displays regressions of an indicator for meeting minimum calorie requirement on log leave-out unit value market rice prices from NSS survey data 2003–2012. All specifications include district-sector-season and period fixed effects. Household controls are log household size, SC/ST, land ownership, religion, cooking fuel, and SES index. Median SES defined using survey weights, so observation counts are different across above- and below-median groups. Household-level SES is the predicted value from a regression of log expenditure per capita on permanent household characteristics, with district-sector-season and period fixed effects. Period fixed effects include year-quarter and NSS round fixed effects. All households owning 0.01 hectares of land or greater are classified as landowning. Standard errors in parentheses and clustered at the region-sector level.

a quality-constant bundle of commodities over time. In particular, we use the RPS price measure for “medium-grade” rice throughout.³²

Column 5 presents results using the baseline price measure (NSS unit values), restricting to the subsample of rural districts where the RPS is conducted. Column 6 presents the baseline specification using the RPS price measure. The point estimates are almost identical and in both cases are statistically significant ($p < 0.01$). The calorie-price sensitivity is also much higher for this subsample: a 10 percent increase in price is associated with a 3.04 percentage point decrease in the likelihood of meeting the MCR. The robustness of our results to using the RPS is reassuring and alleviates concerns about potential bias from measurement error or quality substitution in our unit value measure of local prices.

We next examine heterogeneity in calorie-price sensitivity by demographic categories that are commonly used to target policy: SES status, rural versus urban, and landowning (Table 4). We find that a 10 percent increase in rice prices is associated with a 2.33 percentage point reduction in the likelihood of meeting MCR for below-median-SES households and a 1.90 percentage point reduction for rural households. These effects are statistically significantly larger than those for above-median-SES and urban households, for which the estimates are small and insignificant.³³ Table 4 also reports estimates that divide the rural sample into landless and landowning households. The estimate for landless households is larger in magnitude, but the difference is only statistically significant at the 10 percent level.

We also examine caloric sensitivity by more granular SES categories. Online Appendix Figure A4 reestimates equation (11) with interactions between log price and indicators for households belonging to each of the SES quintiles. Caloric sensitivity decreases substantially throughout the SES distribution; a 10 percent increase

³² https://mospi.gov.in/sites/default/files/publication_reports/Report_TACon%20SPCL_on_BaseRevision_CPI_4mar15.pdf, accessed October 9, 2023.

³³ The effect for rural households is smaller than for the RPS sample in Table 3. This may possibly reflect the fact that RPS data are collected from a fixed set of 603 villages/markets chosen because they are ones that “rural agricultural labourers visit”; see <http://mospi.gov.in/price-collection-survey> for more details.

TABLE 5—LOG CALORIES PER CAPITA AND MARKET PRICES BY SUBSAMPLE

	All (1)	By median SES		By census region		Rural by landowning	
		Below (2)	Above (3)	Rural (4)	Urban (5)	Landless (6)	Landowning (7)
log market rice price	−0.074 (0.031)	−0.130 (0.041)	−0.028 (0.031)	−0.112 (0.042)	−0.016 (0.032)	−0.139 (0.082)	−0.119 (0.035)
Equality of effect (<i>p</i> -value)			0.01		0.07		0.80
Observations	524,911	211,796	313,115	316,234	208,677	63,614	252,620

Notes: This table displays regressions of log calories per capita on log leave-out unit value market rice prices from NSS survey data 2003–2012. All specifications include district-sector-season and period fixed effects. Household controls are log household size, SC/ST, land ownership, religion, cooking fuel, and SES index. Median SES defined using survey weights, so observation counts are different across above- and below-median groups. Household-level SES is the predicted value from a regression of log expenditure per capita on permanent household characteristics, with district-sector-season and period fixed effects. Period fixed effects include year-quarter and NSS round fixed effects. All households owning 0.01 hectares of land or greater are classified as landowning. Standard errors in parentheses and clustered at the region-sector level.

in prices would decrease the likelihood of meeting the MCR by 1.9 percentage points for the poorest households but by only 0.1 percentage points for the upper quintile.

One possible explanation for the observed heterogeneity is that above-median-SES, urban households, and rural landowning households are simply further away from the MCR and therefore have lower sensitivity to falling below this threshold. To distinguish this explanation from underlying differences in caloric sensitivity to prices, we estimate effects using log calories per capita as our outcome variable (Table 5). Our baseline estimate for the full sample implies that a 10 percent increase in the market price is associated with a 0.7 percent reduction in calories per capita ($p < 0.05$, column 1). We again see that the effects are concentrated among below-median-SES and rural households (columns 2 and 4). This cannot be explained by differences in the average levels of prices or variability across the groups: in fact, as shown in Table 2, average prices and the residual standard deviations are nearly the same for above-median-SES and urban households. It is also unlikely to be due to calorie satiation: in the cross section, calories increase with respect to expenditure throughout the expenditure distribution (see Figure 1). In contrast, the coefficients for rural landless and landowning are very similar, suggesting that the higher sensitivity of meeting the MCR for landless households reflects that they are closer to the calorie threshold (columns 6 and 7 of Table 5).

What do these results imply about the costs of price risk to households? On average, our full-sample results indicate that a 10 percent increase in rice prices (about 1 SD of the within-district-sector-season price variation; see Table 2) is associated with 1.31 percentage points fewer households—or approximately 16 million individuals extrapolating from India’s population in our study period—meeting the MCR. However, it is important to note that households near the MCR threshold are not the only ones facing welfare losses from price risk. Our results also indicate a negative calorie-price gradient for poorer households, implying that many households *already* below the MCR (close to half of below-median households) experience further shortfalls below minimum requirements when prices rise. Moreover, these correlations exist despite government welfare programs including the PDS.

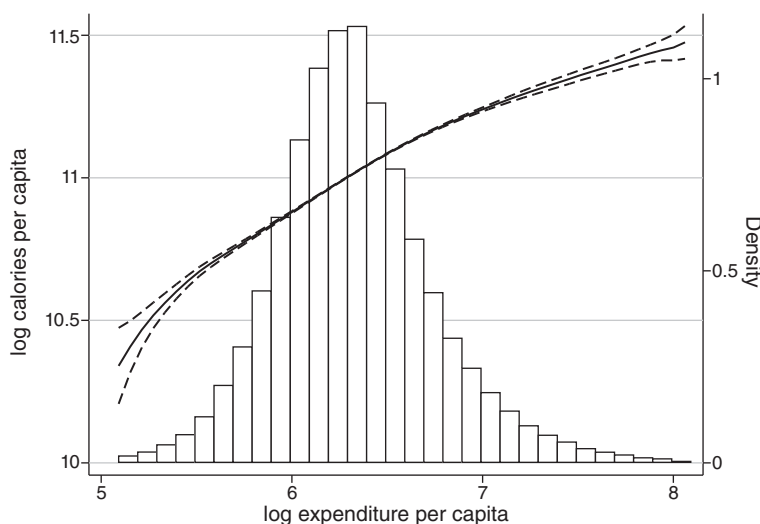


FIGURE 1. LOG CALORIES PER CAPITA VERSUS LOG EXPENDITURE PER CAPITA

Notes: This figure plots a histogram of household log expenditure per capita (right axis) against a line representing a nonparametric regression of log calories per capita on log expenditure per capita (left axis), using data from the National Sample Survey 2003–2012. Regression and histogram both condition on district-sector-quarter fixed effects to nonparametrically adjust for prices. Dashed lines represent 95 percent confidence interval, clustered at the district-sector level.

Finally, as Chetty and Looney (2006) argue, the welfare consequences of risk are likely underestimated given the actions highly risk-averse households take to smooth consumption.³⁴ Taken together, these results suggest substantial losses in welfare from price variability.

In the context of the theory, these results imply that households would benefit from policy tools that provide insurance against price risk, such as indexed transfers that increase with local prices or in-kind transfers. This result is driven by poorer households, precisely those typically targeted by safety net programs.³⁵ Intuitively, the fact that households fall below the MCR in high-price states—despite allowing potential substitution to other goods, price-income correlations, and available smoothing technologies—implies that the marginal utility of a dollar for households is high in these states and that they would value an insurance menu in which transfers increase with respect to price. In-kind transfers provide such this type of insurance (relative to fixed cash transfers) since the value of in-kind transfers rises automatically when local prices rise.

³⁴ Indeed, there is a long tradition of documenting these actions in the context of India, for example, the accumulation of bullocks (Rosenzweig and Wolpin 1993) as well as female migration for marriage (Rosenzweig and Stark 1989).

³⁵ Note that although we do not observe a significant correlation between prices and caloric intake for above-median households, we cannot rule out welfare gains from in-kind relative to a fixed cash transfer. Recall that if households are engaging in costly consumption smoothing behavior, our test provides a sufficient but not necessary condition for in-kind transfers to deliver benefits through an insurance mechanism.

IV. An Evaluation of India's In-Kind Transfer Program

Having demonstrated the benefit to households from in-kind transfers, we next turn to a policy evaluation of India's flagship in-kind transfer program: the PDS. The goal of this analysis is to determine whether the PDS actually targets the households that would benefit from in-kind transfers and the extent to which these transfers mitigate caloric sensitivity to prices. Analyzing the "on the ground" effects of the PDS is particularly relevant given potential problems with targeting, rationing, and leakage (Government of India Planning Commission 2005; Niehaus et al. 2013; Dreze and Khera 2015; Banerjee et al. 2021). In addition, corruption in distribution might increase precisely during high-price periods (Hari 2016). We use policy variation in PDS generosity to instrument for observed PDS value and estimate the causal effects of the PDS on household caloric intake and the sensitivity of calories to prices. We conclude by providing evidence against alternatives to the insurance mechanism.

A. PDS Policy Variation

PDS commodities are procured by the central government and made available to state governments at significantly discounted prices. States purchase these commodities at the discounted price, transport them to a network of shops, and sell them to beneficiaries. The federal government sets minimum guidelines for the program by determining maximum prices at which PDS goods can be sold, minimum entitlements per household, and mandated categories of eligible beneficiaries. However, states can and do use state resources to lower prices further and expand entitlements and eligibility beyond these federal requirements (see Balani 2013 for details on the functioning of the PDS). Therefore, the generosity of the PDS varies across states and over time, and we exploit this source of variation to estimate the causal effects of the PDS. We address potential policy endogeneity in detail below.

There is no comprehensive data source for state PDS policies.³⁶ We therefore construct measures of PDS generosity at the state-year level on both the price and quantity margins as follows. We observe statutory PDS prices in the *Foodgrain Bulletin*, an annual government report (Department of Food and Public Distribution 2003–2012).³⁷ The *Bulletin* is not comprehensive, so we additionally surveyed newspaper databases to identify other policy changes and to get more exact information on the date of *Bulletin* price changes. Combined, we have as complete a dataset of PDS price changes as is possible.

The quantity component of PDS value reflects both the number of eligible households and quotas per eligible household. However, there is no consistent source of information on changes to either. To identify policy changes in eligibility, we use NSS data to find sharp breaks in observed PDS value received by households and then check in newspapers and state records to see if there was a policy change at that time. Specifically, we simulated potential policy changes for each state s and

³⁶For PDS policy changes in select states, see Khera (2011).

³⁷When different card types are charged different prices, we use the BPL price in all calculations. This is by necessity—our data do not list card type—but the vast majority of households using the PDS pay BPL prices (Niehaus et al. 2013).

year-quarter t in the following way. We ran regressions of PDS value on state and time fixed effects, controls for household characteristics and known policies, and an indicator for being in state s after time t . Whenever the coefficient on the indicator was larger than ₹10 in absolute value, we checked newspapers and state records. If we found explicit, credible mention of an increase in eligibility, we coded that period as an eligibility increase for the given state. We find five such eligibility increases, which we list in online Appendix Table A6. Changes in quotas for eligible households are often small and ad hoc and difficult to identify cleanly in the data. We therefore do not exploit this source of variation in our quantity instrument. We provide more information on the political context in which states' PDS policies changed as well as study the determinants of reform in online Appendix A3.

The generosity of the PDS as observed in the NSS increased dramatically over the study period. Panel A of Figure 2 shows that average real PDS rice prices more than halved over our study period, from over ₹5 to ₹2. Panel B shows that while quantities for beneficiaries stayed roughly constant, the number of beneficiaries doubled from 20 percent to 38 percent of households. This translated into a 300 percent increase in the value of the PDS subsidy over the period (panel C), from ₹16 to ₹47 (average across all households). Online Appendix Figure A1 plots the share of households consuming PDS rice against per capita expenditure, deflated to the 1999 price level. We display this relationship for 2008 and earlier, before most of the big expansions in eligibility, and for 2009 and after. Households became more likely to access the PDS at all expenditure levels over time, but the gains were most pronounced for very poor households.

B. PDS Value and Instruments

We calculate the subsidy value v_{irt} for each household using information on the leave-out market prices $p_{-i,rt}$, PDS prices $p_{-i,rt}^{PDS}$, and observed PDS consumption q_{it}^{PDS} .³⁸ The value of the PDS rice subsidy can be written as

$$v_{irt} = (p_{-i,rt} - p_{-i,rt}^{PDS}) q_{it}^{PDS}.$$

Differences between market prices and PDS prices are substantial, leading to a large transfer to households. The average price for PDS rice was ₹3.5 per kilogram, compared to a market price of ₹9.9. In our sample, the average monthly transfer adds up to ₹109 for rice beneficiaries (conditional on obtaining PDS rice), 4.9 percent of the ₹2,205 average monthly expenditure. This is likely the single largest government transfer for most households: in comparison, transfers from the National Rural Employment Guarantee Scheme (NREGS), India's other major social welfare program, made up only 1.8 percent of beneficiaries' expenditure in Andhra Pradesh in 2012 (Muralidharan, Niehaus, and Sukhtankar 2017b).

To isolate changes in PDS value driven by policy changes, we instrument PDS value by changes in states' PDS policies. Our first instrument is simply p_{st}^{BPL} , the

³⁸ As discussed above, we define market prices and PDS prices by the mean region-sector-year-quarter leave-out unit values. The market unit value is based on the 88.3 percent of households that consume rice from the market; the PDS unit value is based on the 25.7 percent of households that consume rice from the PDS.

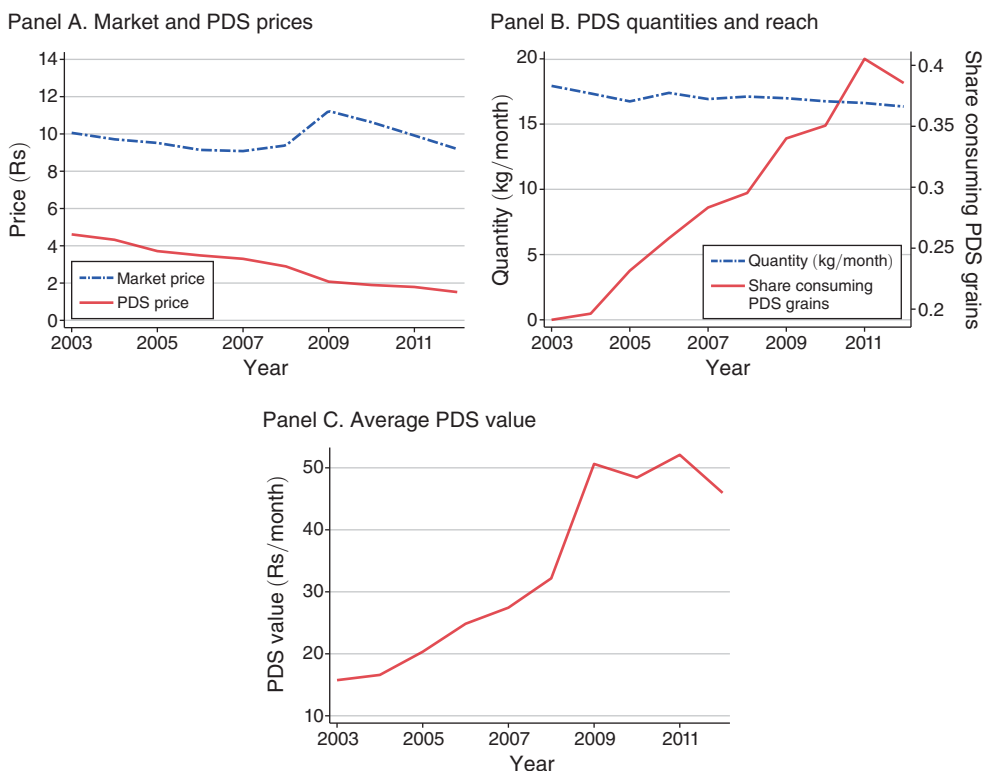


FIGURE 2. PDS GENEROSITY AND COVERAGE OVER TIME

Notes: This figure shows the evolution of benefit generosity in the PDS using data from the National Sample Survey 2003–2012. Panel A shows market and PDS mean unit values over time. Panel B shows PDS quantities for beneficiaries and the total share of households who consume PDS goods. Panel C shows unconditional average monthly PDS generosity $\mathbb{E}[(p_{-i,rt}^{mkt} - p_{-i,rt}^{PDS})q_{it}]$. All units are deflated to 1999 rupees, which traded at 43 to 1 with the US dollar.

statutory PDS price of rice charged to families classified as BPL—the vast majority of PDS beneficiaries—in state s at time t . Panel A of online Appendix Figure A2 shows a particularly striking example of changes in PDS prices, when Andhra Pradesh lowered the PDS rice price from ₹5 to ₹2 in 2008 and then to Re1 four years later.

Our second instrument is an indicator E_{ist} equal to 1 if household i is in a state s in which a major PDS eligibility increase has occurred prior to the survey date. Panel B of online Appendix Figure A2 shows the importance of additionally accounting for this variation, highlighting the increase in PDS value when Odisha universalized the PDS in a poor region of the state, approximately doubling PDS participation in one year.

C. Empirical Strategy

We examine the direct effect of PDS generosity on caloric outcomes as well as the effect of the PDS on the sensitivity of calories to market prices. Our first estimating equation is

$$(12) \quad c_{ist} = \alpha_1 v_{irt} + \alpha_2 p_{-i,rt} + X_{it}\lambda + \delta_{da} + \tau_t + \phi_{round} + \varepsilon_{ist},$$

where s indexes states (other indices as previously defined), c_{ist} is our calorie outcome measure, $p_{-i,rt}$ is the leave-out market price, and α_1 is the coefficient of interest. Standard errors are clustered at the state level, which is the level of PDS policy variation. All regressions are estimated using NSS weights. We instrument for observed PDS value v_{irt} with three instruments: the statutory PDS price at the time the household was surveyed, an indicator for whether the household was surveyed after a major eligibility expansion in its state, and the interaction between the two.³⁹

To determine the effect of the PDS on caloric sensitivity to market prices, we estimate

$$(13) \quad c_{ist} = \beta_1 p_{-i,rt} + \beta_2 p_{-i,rt} \times v_{irt} + \beta_3 v_{irt} + X_{it} \lambda + \delta_{da} + \tau_t + \phi_{round} + \varepsilon_{ist}.$$

We instrument for v_{irt} as above and for $p_{-i,rt} \times v_{irt}$ using our three instruments as well as their interactions with the market price. Our main coefficient of interest is β_2 , which is identified by comparing the correlation between rice prices and calorie outcomes at different levels of instrumented PDS generosity.

The key identifying assumption is that policy changes in PDS generosity are not endogenous to local conditions or correlated with other unobserved changes that might affect calories or calorie-price sensitivity directly. For example, we might be concerned that expansions of the PDS occur during good economic times or in response to calorie shortfalls.

To assess this possibility, we estimate event-study specifications for the dynamic effect of the eligibility expansion instrument on PDS transfer value and meeting the MCR.⁴⁰ We use an imputation approach that uses only pre- and never-treated observations to predict counterfactual nontreated outcomes (Borusyak, Jaravel, and Spiess 2021), although our results look similar if we use a conventional two-way fixed effects specification (see online Appendix Figure A6).

Figure 3 displays the results, with the first-stage effect of the expansion on PDS value in panel A and the reduced-form effect on meeting the MCR in panel B. We see no differential trends in average PDS value in the years before the reform. However, PDS value v_{irt} begins to increase immediately following the reforms. Within 5 years, v_{irt} increases by ₹50 on average, approximately 50 percent of the mean PDS transfer received by beneficiaries during our study period. Panel B of Figure 3 also provides no evidence of changes in whether households meet the MCR before a policy is implemented—supporting the parallel trends assumption—but a sharp increase after PDS expansion, indicating that the PDS substantially improved caloric consumption.

³⁹de Chaisemartin and D'Haultfœuille (2020) decompose the two-way fixed effect estimand into a weighted average of treated area-period-specific effects and point out that since those weights may be negative, under heterogeneous treatment effects, the conventional estimand may be opposite-signed from the group-specific average treatment on treated effects. We calculate these weights in the differences-in-differences estimate of the effect of the expansions on outcomes in online Appendix Figure A3 (the theory to compute these weights for continuous regressors does not yet exist) and find that all but 13 of the 2,756 treated district-sector-quarters have positive weights.

⁴⁰With small and frequent changes to PDS prices, our price instrument is not conducive to this type of graph; we show below that results go through with the expansion instrument only.

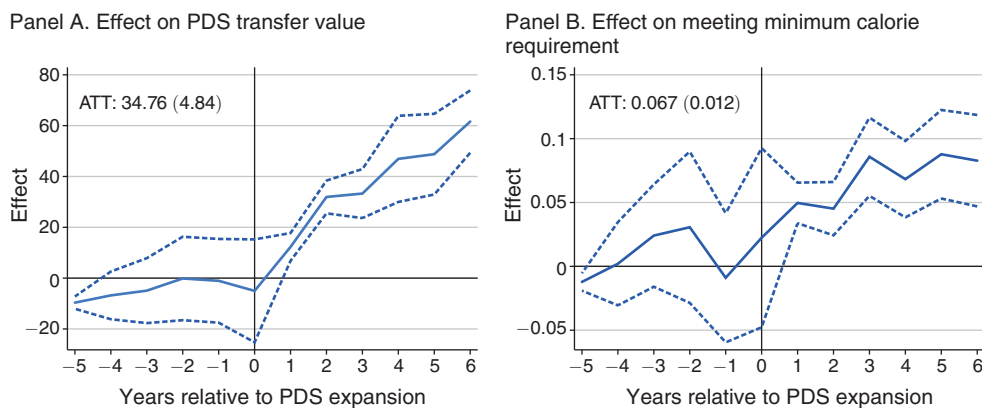


FIGURE 3. EFFECT OF PDS ELIGIBILITY EXPANSIONS ON PDS TRANSFER VALUE AND CALORIC INTAKE

Notes: This figure shows event study coefficients from a regression of the outcome (PDS value in panel A and an indicator for whether the household meets minimum calorie requirements in panel B) on time relative to policy expansion: $y_{idt} = \sum_{\tau} \beta_{\tau} \mathbf{1}_{\tau} + X_{idt} \alpha + \gamma_d + \varphi_t + \varepsilon_{iat}$, for household i in district-sector-season d and year t at year relative to expansion τ , where controls include PDS rice price, log household size, SC/ST, land ownership, religion, cooking fuel, and SES index. Household-level SES is the predicted value from a regression of log expenditure per capita on permanent household characteristics, with district-sector-season, year-quarter, and NSS round fixed effects. Models are estimated using the imputation approach of Borusyak, Jaravel, and Spiess (2021). Standard errors clustered at the state level.

D. Results

Table 6 contains first-stage results for equation (12) for the overall sample and separately for below- and above-median-SES households (online Appendix Table A7 contains the first stage for all demographic subgroups). We present throughout the robust effective F -statistic (Montiel Olea and Pflueger 2013), which can accommodate both heteroskedasticity and multiple excluded instruments, and report the critical values for when the worst-case bias of two-stage least squares exceeds 10 percent of the worst-case bias of OLS.⁴¹ The coefficients for PDS price decreases and eligibility increases are both strongly significant and have the expected signs. In the full sample, reducing the government-mandated BPL price by Re1 increases the value of the PDS transfer by ₹9.7. On average, our increases in eligibility increase the value of the PDS by ₹51/month, 2 percent of average total household expenditure. These coefficients are broadly similar and again strongly significant in the below- and above-median subsamples.

Panel A of Table 7 presents our results on the effects of PDS generosity on the likelihood a household meets the MCR. An increase of ₹100 in PDS value leads to a

⁴¹ Our effective F -stats are above the critical values in the main sample and slightly under these critical values in the subsamples. We note that in a setting like ours with multiple excluded instruments, these effective F -statistics could be compared to either a value of ten or the Montiel Olea and Pflueger (2013) critical values we report (Andrews, Stock, and Sun 2019). In practice, both the critical values and the effective F -stats are always clearly above ten, so reporting these critical values is the conservative approach. Unfortunately, there is no method to produce weak-instrument robust F -stats in the multiple endogenous regressor case in Tables 7 and 8, panel B—see Andrews, Stock, and Sun (2019) for a discussion. However, the results for the single endogenous regressor case are reassuring and suggest that our instruments in the multiple endogenous regressors case are likely to be of similar strength.

TABLE 6—FIRST STAGE OF PDS VALUE (IN 100s OF RUPEES) ON INSTRUMENTS

	All (1)	Below-median SES (2)	Above-median SES (3)
PDS price (₹)	−0.097 (0.035)	−0.126 (0.043)	−0.064 (0.025)
Eligibility increase (=1)	0.513 (0.103)	0.542 (0.114)	0.449 (0.093)
Eligibility increase × PDS price	−0.116 (0.038)	−0.099 (0.049)	−0.120 (0.025)
Effective <i>F</i> -stat	19.07	17.03	16.37
10 percent bias crit. val.	18.66	17.84	18.87
Observations	524,911	211,796	313,115

Notes: This table reports regressions of PDS transfer value on PDS statutory rice prices, PDS expansion indicator, and their interaction. PDS value is calculated as the difference between market and PDS rice prices multiplied by household-level PDS quantities (expressed in units of 100). Market and PDS prices are average unit values of market and PDS rice at region-sector-period level. Statutory rice prices are state-mandated prices per kilogram of PDS rice for households below the poverty line. Expansion indicates if a household is surveyed in an expansion state after the date of expansion of the PDS reported in online Appendix Table A6. All prices are deflated to 1999 rupees. All specifications include district-sector-season and period (calendar quarter and NSS round) fixed effects. Household controls include log market rice unit value, log household size, SC/ST, land ownership, religion, cooking fuel, and SES index. Household-level SES is the predicted value from a regression of log expenditure per capita on permanent household characteristics, with district-sector-season, year-quarter, and NSS round fixed effects. Standard errors in parentheses and clustered at the state level.

10.7 percentage point increase in the likelihood that the household meets the MCR (column 1). Panel C of Figure 2 shows that PDS value increased by ₹30.1 over the study period; extrapolating to the entire population implies that the expansions in PDS generosity increased the number of people meeting the MCR by just over 40 million. Calories per capita also increase by 6 percent for every ₹100 of PDS value (column 4 of Table 7; $p = 0.121$). We find a positive and significant effect on the likelihood of meeting the MCR for below-median households and a positive but insignificant effect for above-median households. The point estimates for log calories per capita are very similar (but insignificant) for both subgroups. These results suggest that an increase in PDS has a similar effect on calories for the two groups (consistent with the relatively constant calorie-income gradient we document in Figure 1) but a larger effect on meeting the MCR for below-median households who are on average more likely to be just below the minimum calorie threshold.⁴²

Panel B of Table 7 demonstrates that expansions in PDS generosity also decrease household sensitivity to market prices. The first row shows the implied effect of an increase in the market price for a household without any (instrumented) PDS consumption; the second row shows the interaction of market price and a ₹100 increase in PDS value. We also provide the predicted effect of market rice price at the mean PDS value. A 10 percent increase in prices for a household without any (instrumented) PDS consumption decreases the likelihood the household meets the MCR by 2.6 percentage points (column 1). However, every ₹100 of PDS value reduces this drop by 1.8 percentage points. Our results imply that households' caloric intake would no longer be sensitive to market prices if they received a ₹147 transfer from

⁴²Note, however, that the complier population (those for whom increases in policy generosity lead to on-the-ground increases in PDS value) are different across these subgroups.

TABLE 7—EFFECT OF PDS GENEROSITY ON CALORIC OUTCOMES

	Meets MCR			log calories per capita		
	All	Below- median SES	Above- median SES	All	Below- median SES	Above- median SES
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A. IV of outcomes on PDS value</i>						
PDS value (100 ₹)	0.107 (0.052)	0.136 (0.063)	0.078 (0.048)	0.064 (0.040)	0.062 (0.039)	0.069 (0.041)
Equality of effects (<i>p</i> -value)			0.049			0.690
Effective <i>F</i> -stat	19.07	17.03	16.37	19.07	17.03	16.37
10 percent bias crit. val.	18.66	17.84	18.87	18.65	18.15	18.85
<i>Panel B. IV of outcomes on PDS value</i>						
log market rice price	−0.260 (0.054)	−0.468 (0.086)	−0.123 (0.044)	−0.166 (0.033)	−0.260 (0.057)	−0.106 (0.031)
Market rice price × PDS value	0.177 (0.066)	0.209 (0.075)	0.274 (0.105)	0.150 (0.049)	0.143 (0.045)	0.247 (0.073)
Equality of effects (<i>p</i> -value)						
log market rice price			0.000			0.011
Market rice price × PDS value			0.438			0.027
Pred. rice elasticity at mean PDS	−0.207 (0.051)	−0.384 (0.094)	−0.071 (0.036)	−0.122 (0.033)	−0.203 (0.058)	−0.059 (0.027)
Mean PDS value	0.30	0.40	0.19	0.30	0.40	0.19
SD PDS value	0.604	0.668	0.512	0.604	0.668	0.512
First percentile PDS value	0.000	0.000	0.000	0.000	0.000	0.000
Ninety-ninth percentile PDS value	2.556	2.685	2.325	2.556	2.685	2.325
Observations	524,911	211,796	313,115	524,911	211,796	313,115

Notes: This table shows coefficients from regressions of an indicator for meeting the minimum calorie requirement (MCR, columns 1 and 2) or log calories per capita (columns 3 and 4) on PDS value (in panel A) and PDS value, market rice prices, and their interaction (panel B). In panel A, PDS value is calculated as the difference between market and PDS rice prices multiplied by household-level PDS quantities (expressed in units of 100 rupees), and instrumented for with state-level statutory PDS prices, a dummy for state-level PDS expansions, and their interaction. In panel B, the same three instruments are included, as well as their interactions with market prices. For comparison, mean per capita expenditure is ₹711. “Pred. rice elasticity” is taken at mean PDS value. All specifications include district-sector-season and period (calendar quarter and NSS round) fixed effects. Controls include log market rice unit value, log household size, SC/ST, land ownership, religion, cooking fuel, and SES index. Effective *F*-stat calculated using Montiel Olea and Pflueger (2013). Standard errors clustered at the state level in parentheses.

the PDS, which is roughly one-half larger than the average nonzero transfer. We observe similar patterns when we use log calories per capita as the outcome measure (column 4).

The remaining columns show the results separately for below- and above-median-SES households. We see larger calorie-price sensitivity at mean PDS levels for below-median households (consistent with our results in Tables 4 and 5), but interestingly, we see that both below- and above-median households would experience caloric sensitivity to prices at zero (imputed) PDS. For both groups, higher PDS transfers reduce price sensitivity. These results imply that the PDS as implemented during the study period provided households with substantial insurance against price risk, both for below- and above-median households.

In online Appendix Tables A7 and A8, we show results for all subgroups. In brief, the PDS increases caloric intake and reduces sensitivity to market prices for all subsamples and consistent with Table 4, reduces sensitivity of urban and rich households to basically zero.

TABLE 8—EFFECT OF PDS GENEROSITY ON MEETING MINIMUM CALORIE REQUIREMENT

	All				Below-median SES			
	No suppliers (1)	Pol. econ. controls (2)	Price inst. only (3)	Expansion inst. only (4)	No suppliers (5)	Pol. econ. controls (6)	Price inst. only (7)	Expansion inst. only (8)
<i>Panel A. IV of meeting minimum calorie requirement on PDS value</i>								
PDS value (100 ₹)	0.170 (0.053)	0.109 (0.053)	0.017 (0.085)	0.171 (0.068)	0.224 (0.066)	0.136 (0.065)	0.058 (0.078)	0.198 (0.090)
MOP <i>F</i> -stat	25.07	19.12	8.94	17.44	20.55	16.79	9.07	16.74
10 percent bias crit. val.	21.05	18.57	23.11	23.11	19.60	17.80	23.11	23.11
5 percent bias crit. val.	34.58	30.21	37.42	37.42	32.23	28.97	37.42	37.42
<i>Panel B. IV of meeting minimum calorie requirement on PDS value</i>								
log market rice price	−0.253 (0.047)	−0.269 (0.053)	−0.246 (0.074)	−0.298 (0.070)	−0.442 (0.074)	−0.466 (0.084)	−0.473 (0.093)	−0.488 (0.108)
Market price × PDS value	0.171 (0.119)	0.191 (0.071)	0.256 (0.112)	0.129 (0.074)	0.163 (0.117)	0.202 (0.073)	0.322 (0.117)	0.128 (0.082)
Predicted rice elasticity, at mean PDS value	−0.205 (0.057)	−0.212 (0.052)	−0.170 (0.063)	−0.260 (0.066)	−0.380 (0.106)	−0.385 (0.091)	−0.344 (0.094)	−0.436 (0.117)
Mean PDS value	0.28	0.30	0.30	0.30	0.38	0.40	0.40	0.40
SD PDS value	0.603	0.606	0.606	0.606	0.669	0.670	0.670	0.670
First percentile PDS value	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Ninety-ninth percentile PDS value	2.581	2.559	2.559	2.559	2.698	2.700	2.700	2.700
Observations	391,176	524,911	524,911	524,911	160,154	211,772	211,772	211,772

Notes: This table shows coefficients from regression of a dummy for meeting the minimum caloric requirement on PDS value (in panel A) and PDS value, market rice prices, and their interaction (panel B). In panel A, PDS value is calculated as the difference between market and PDS rice prices multiplied by household-level PDS quantities (expressed in units of 100 rupees) and instrumented for with state-level statutory PDS prices, a dummy for state-level PDS expansions, and their interaction. In panel B, the same three instruments are included as well as their interactions with market prices. Model (1) includes all PDS instruments, (2) includes all PDS instruments but excludes states supplying the majority of rice to the PDS, (3) includes all PDS instruments but controls for active NREGS program in district at the time of surveying as well as elections at the state-quarter level, (4) instruments for PDS value with statutory rice price instruments alone, and (5) instruments for PDS value with expansion instruments alone. For comparison, mean per capita expenditure is ₹711. All specifications include district-sector-season and period (calendar quarter and NSS round) fixed effects. Controls include log market rice unit value, log household size, SC/ST, land ownership, religion, cooking fuel, and SES index. Weak IV *F*-stats are calculated using Kleibergen-Paap (2006). Standard errors clustered at the state level in parentheses.

E. Robustness

In Table 8, we find that these results are robust to various alternative choices of samples and specifications. First, restricting the sample to only those states that are not major suppliers of rice to the PDS makes no qualitative difference to the results; the coefficients are very similar, suggesting that the results are not driven by procurement or unobserved positive shocks to supply. Next, adding controls related to election cycles as well as the rollout of the NREGS, the other big social welfare program, makes no observable difference to either coefficients or statistical significance. We continue to see mitigating effects of PDS value on caloric sensitivity to prices when we instrument for PDS value using policy variation in prices alone (column 3) or eligibility expansions alone (column 4).⁴³ In online Appendix Table A9,

⁴³We note that results obtained using only one source of variation at a time have low effective first-stage *F*-statistics and should be interpreted with caution.

we include wild cluster bootstrap p -values at the state level and find little change in our inference on the effect of the PDS on caloric sensitivity (panel B).

F. Mechanisms

The decline in caloric sensitivity to prices resulting from expansions in in-kind transfers is consistent with the insurance mechanism posited in the model. We now turn to investigating other mechanisms through which expansions in in-kind transfers could potentially affect household responses to price risk.

First, caloric price sensitivity might decline after PDS expansions merely because households are richer or face lower liquidity constraints. While we know of no experimental or quasi-experimental research on the effect of unconditional cash transfers on price sensitivity, we exploit cross-sectional variation in a measure of permanent income in our sample to estimate the observational effect of income on price-calorie sensitivity. These estimates will provide an upper bound on the causal effect of the income channel on price sensitivity if higher-income households have access to better smoothing technologies, as seems likely.

Specifically, we use the exponent of predicted log consumption per capita from our SES measure as a rupee-denominated measure of permanent income and add an interaction with prices to our baseline regression of meeting the MCR on prices in equation (12). We find that caloric sensitivity to prices declines by 0.013 per additional ₹100 of permanent income, consistent with our results on above- and below-median households in Section IIIB. However, this gradient is relatively small compared to the causal effect of the PDS, where an additional ₹100 of PDS value decreases caloric sensitivity by 0.177. This suggests that the income channel accounts for only a small share of the observed effect.⁴⁴

Second, households could exhibit flypaper effects in spending from in-kind transfers (Hastings and Shapiro 2018), resulting in a higher marginal propensity to consume calories from the PDS than from an equivalent cash transfer. Households with higher caloric intake could in turn exhibit reduced caloric sensitivity to prices. To address this concern, we compare our estimates to Weaver et al. (2023), who provide randomized cash transfers to households in Jharkhand, a state in northeast India. They find a log per capita calorie impact of a ₹500 per month transfer of 0.084. Scaling and normalizing to 1999 rupees gives an estimate of 0.059, extremely close to our estimate of 0.064 (Table 7), indicating a negligible role for flypaper effects in this context.⁴⁵

A potential caveat to the cash transfer estimates is that the study included a mild framing toward maternal and child nutrition. In practice, this took the form of phone call reminders that connected with less than 50 percent of the group; moreover, nutrition impacts on those with phones and without phones are not statistically distinguishable. Nonetheless, for added reassurance, we also perform a

⁴⁴In online Appendix Figure A4, we more flexibly estimate the relationship between permanent income and price elasticities by interacting prices with indicators for the five quintile groups. Price elasticities are higher for poorer groups and statistically significant for the bottom three quintiles.

⁴⁵GDP per capita in the Weaver et al. (2023) sample and our (inflation-adjusted) all-India sample are very comparable: ₹79,873 (Jharkhand) and ₹82,979 (all-India). Sources: <https://mospi.gov.in/data>; https://finance.jharkhand.gov.in/pdf/Economic_Survey_2020_21/Jharkhand_Economic_Survey_2020_21.pdf (accessed July 6, 2023).

similar test to the income effect test above, estimating the cross-sectional gradient of calorie-price sensitivity to a measure of predicted caloric intake based on the same fixed household characteristics we use to construct our SES measure. An additional log point of predicted calories increases the coefficient on log market price by 0.511 (online Appendix Table A12). Multiplying this gradient with the estimates of the effect of ₹100 of PDS transfer on caloric intake (0.064; see panel A of Table 7), we find an implied PDS effect on price sensitivity of 0.033, roughly a fifth of our baseline estimate (0.177; panel B of Table 7). This implies that the increase in the level of caloric intake arising from PDS expansions (even if partially driven by fly-paper effects) is too small to account for the magnitude of reduction in calorie-price sensitivity we observe.

Third, in-kind transfers could have general equilibrium effects on local market prices, as found by Cunha, De Giorgi, and Jayachandran (2018) in a different context. If caloric sensitivity to prices is nonlinear with respect to the market price, such general equilibrium price effects could theoretically affect observed caloric smoothing. We address this possibility in online Appendix Table A10, where we regress market rice prices on instrumented PDS value. Across specifications, we find very small effects of the PDS on prices. Using our baseline set of instruments, we find that an additional ₹100 of PDS generosity decreases market prices by an insignificant 0.6 percent. Given the negligible effect of expansions on market prices, these effects cannot explain the smoothing effects.

These muted price effects are consistent with evidence in Shrinivas, Baylis, and Crost (2019), who also find no effect of PDS expansions on market prices. This difference with Cunha, De Giorgi, and Jayachandran (2018) may be because the transfers they study are much larger than the PDS transfers. Rescaling their estimates implies a decrease of market price of only 1.6 percent per ₹100 of PDS, broadly comparable to our estimate.⁴⁶

Finally, increased PDS generosity could theoretically result in a reduction in the variation of market prices, which could reduce sensitivity to observed price variation if, for example, households are better able to smooth small price shocks than large price shocks. However, we find that the effects of PDS expansions on price variability are small and insignificant (online Appendix Figure A5).

V. Conclusion

We show that in a world in which households are exposed to commodity price risk—a common situation in many developing countries due to poor market integration—*inframarginal* in-kind transfers will be welfare improving relative to cash transfers from the household perspective if and only if the marginal utility of income is increasing with respect to price. Intuitively, in-kind transfers provide insurance since the value of the transfer rises automatically with the price of the transferred good. Testing this condition empirically in the context of India, we find that in-kind transfers provide insurance value to households in addition to pure redistribution.

⁴⁶A ₹100 transfer represents only 3.5 percent of expenditure, while the transfers in Cunha, De Giorgi, and Jayachandran (2018) were 10 percent of expenditures. Rescaling their 4 percent estimate implies a $(10/3.5) \times 4\% = 1.6\%$ effect.

In addition, we provide the first evidence that in-kind transfers do in fact smooth household outcomes in the face of price fluctuations, demonstrating that expansions of the Public Distribution System not only increase caloric intake by households but also reduce sensitivity of calories to local prices.

These results have important implications in the context of ongoing global debates about the optimal form of social protection programs. In particular, a recent and growing body of evidence documenting the success of unconditional cash transfers has garnered much media and policy attention and influenced the ways in which donors choose to allocate funds.⁴⁷

We stress that our results do not imply that in-kind transfers necessarily dominate fixed cash transfers from a social planner perspective: a full welfare analysis would need to take into account the social cost of provision, including potential differences in implementation costs (see, e.g., Banerjee et al. 2021; Gentilini 2016; Margolies and Hoddinott 2015). Nevertheless, they elucidate an important advantage of in-kind transfers that should be taken into consideration in the design of social protection programs as well as a possible explanation for why beneficiaries themselves might value in-kind transfers.

It is important to note that the relative benefits of in-kind versus cash will vary geographically and over time, based on differences and changes in underlying market integration and resulting price volatility. In addition, this potential benefit of in-kind transfers—mitigation of exposure to price risk—may be difficult to capture in existing randomized controlled trials, which generally measure (relatively) short-run outcomes. We see this as an important area for future research, and a key advantage of the welfare test we propose is that it does not require exogenous variation in prices and can therefore be applied in a variety of settings.

More broadly, our results speak to the importance of considering household exposure to price risk in the design of safety net programs. While in-kind transfers are one way to provide insurance, they are not the only policy instrument that could improve welfare in the presence of price risk. For example, targeting rules for cash transfers may want to take into account local geographic price indicators, such as the average level of staple commodity prices, historical levels of price volatility, or proxies for household ability to smooth price variation. In addition, improvements in digital technology are rapidly changing the landscape for decentralized information collection, opening the possibility for first-best price-indexed cash transfers. We hope that our paper serves as a starting point for further work in this important area.

REFERENCES

- Ahmed, Akhter, John F. Hoddinott, Shalini Roy, Esha Sraboni, Wahidur R. Quabili, and Amy Margolies. 2022. *Which Kinds of Social Safety Net Transfers Work Best for the Ultra Poor in Bangladesh? Operation and Impacts of the Transfer Modality Research Initiative*. Washington, DC: International Food Policy Research Institute.
- Allen, Treb. 2014. “Information Frictions in Trade.” *Econometrica* 82 (6): 2041–83.
- Andrews, Isaiah, James H. Stock, and Liyang Sun. 2019. “Weak Instruments in Instrumental Variables Regression: Theory and Practice.” *Annual Review of Economics* 11 (1): 727–53.

⁴⁷ See, for example, <https://www.poverty-action.org/impact/cash-transfers-changing-debate-giving-cash-poor>, accessed February 12, 2021.

- Atkin, David. 2013. "Trade, Tastes, and Nutrition in India." *American Economic Review* 103 (5): 1629–63.
- Atkin, David, and Dave Donaldson. 2015. "Who's Getting Globalized? The Size and Implications of Intra-National Trade Costs." NBER Working Paper 21439.
- Attanasio, Orazio, Vincenzo Di Maro, Valérie Lechene, and David Phillips. 2013. "Welfare Consequences of Food Prices Increases: Evidence from Rural Mexico." *Journal of Development Economics* 104: 136–51.
- Balani, Sakshi. 2013. "Functioning of the Public Distribution System: An Analytical Report." Unpublished.
- Banerjee, Abhijit, Paul Niehaus, and Tavneet Suri. 2019. "Universal Basic Income in the Developing World." *Annual Review of Economics* 11: 959–83.
- Banerjee, Abhijit, Rema Hanna, Benjamin A. Olken, Elan Satriawan, and Sudarno Sumarto. 2021. "Food vs. Food Stamps: Evidence from an At-Scale Experiment in Indonesia." NBER Working Paper 28641.
- Barrett, Christopher B. 1996. "On Price Risk and the Inverse Farm Size-Productivity Relationship." *Journal of Development Economics* 51 (2): 193–215.
- Barrett, Christopher B. 2002. "Food Security and Food Assistance Programs." In *Handbook of Agricultural Economics*, Vol. 2, edited by Bruce L. Gardner and Gordon C. Rausser, 2103–90. Amsterdam: Elsevier.
- Batista, Catia, Dan Silverman, and Dean Yang. 2015. "Directed Giving: Evidence from an Inter-Household Transfer Experiment." *Journal of Economic Behavior & Organization* 118: 2–21.
- Behrman, Jere R., Anil B. Deolalikar, and Barbara L. Wolfe. 1988. "Nutrients: Impacts and Determinants." *World Bank Economic Review* 2 (3): 299–320.
- Bellemare, Marc F., and Yu Na Lee. 2016. "Attitudes to Price Risk and Uncertainty: The Earnest Search for Identification and Policy Relevance." *Canadian Journal of Agricultural Economics* 64 (4): 599–612.
- Bellemare, Marc F., Christopher B. Barrett, and David R. Just. 2013. "The Welfare Impacts of Commodity Price Volatility: Evidence from Rural Ethiopia." *American Journal of Agricultural Economics* 95 (4): 877–99.
- Besley, Timothy, and Stephen Coate. 1991. "Public Provision of Private Goods and the Redistribution of Income." *American Economic Review* 81 (4): 979–84.
- Blattman, Chris, Michael Faye, Dean Karlan, Paul Niehaus, and Chris Udry. 2017. "Cash as Capital." *Stanford Social Innovation Review* 15 (3): 57–58.
- Borusyak, Kirill, Xavier Jaravel, and Jann Spiess. 2021. "Revisiting Event Study Designs: Robust and Efficient Estimation." Unpublished.
- Brinkman, Henk-Jan, Saskia De Pee, Issa Sanogo, Ludovic Subran, and Martin W. Bloem. 2010. "High Food Prices and the Global Financial Crisis have Reduced Access to Nutritious Food and Worsened Nutritional Status and Health." *Journal of Nutrition* 140 (1): 153S–61S.
- Brown, Caitlin, Rossella Calvi, and Jacob Penglase. 2018. "Sharing the Pie: Undernutrition, Intra-household Allocation, and Poverty." Unpublished.
- Burke, Marshall, Lauren Falcao Bergquist, and Edward Miguel. 2019. "Sell Low and Buy High: Arbitrage and Local Price effects in Kenyan Markets." *Quarterly Journal of Economics* 134 (2): 785–842.
- Casaburi, Lorenzo, Rachel Glennerster, and Tavneet Suri. 2013. "Rural Roads and Intermediated Trade: Regression Discontinuity Evidence from Sierra Leone." Unpublished.
- Chetty, Raj. 2009. "Sufficient Statistics for Welfare Analysis: A Bridge Between Structural and Reduced Form Methods." *Annual Review of Economics* 1 (1): 451–88.
- Chetty, Raj, and Adam Looney. 2006. "Consumption Smoothing and the Welfare Consequences of Social Insurance in Developing Economies." *Journal of Public Economics* 90 (12): 2351–56.
- Coate, Stephen. 1989. "Cash Versus Direct Food Relief." *Journal of Development Economics* 30 (2): 199–224.
- Coate, Stephen, Stephen Johnson, and Richard Zeckhauser. 1994. "Pecuniary Redistribution Through In-Kind Programs." *Journal of Public Economics* 55 (1): 19–40.
- Cunha, Jesse M. 2014. "Testing Paternalism: Cash versus In-Kind Transfers." *American Economic Journal: Applied Economics* 6 (2): 195–230.
- Cunha, Jesse M., Giacomo De Giorgi, and Seema Jayachandran. 2018. "The Price Effects of Cash Versus In-Kind Transfers." *Review of Economic Studies* 86 (1): 240–81.
- Currie, Janet, and Douglas Almond. 2011. "Human Capital Development Before Age Five." In *Handbook of Labor Economics*, Vol. 4, edited by Orley Ashenfelter and David Card, 1315–1486. Amsterdam: Elsevier.

- Currie, Janet, and Firouz Gahvari. 2008. "Transfers in Cash and In-Kind: Theory Meets the Data." *Journal of Economic Literature* 46 (2): 333–83.
- Das, Aurolipsa, and Narayan Sethi. Forthcoming. "Cash and In-Kind Transfers in India: Contexts, Preferences and Evidence." *International Journal of Social Economics*.
- Deaton, A., and A. Tarozzi. 2005. "Prices and Poverty in India." In *The Great Indian Poverty Debate*. New Delhi: MacMillan.
- Deaton, Angus. 1989. "Rice Prices and Income Distribution in Thailand: A Non-parametric Analysis." *Economic Journal* 99 (395): 1–37.
- de Chaisemartin, Clément, and Xavier D'Haultfœuille. 2020. "Two-Way Fixed Effects Estimators with Heterogeneous Treatment Effects." *American Economic Review* 110 (9): 2964–96.
- Department of Food and Public Distribution. 2003–2012. *Food Grain Bulletins*. Government of India.
- Department of Planning-cum-Finance. 2021. *Jharkhand Economic Survey 2020–21*. Government of Jharkhand.
- Deschamps, Robert. 1973. "Risk Aversion and Demand Functions." *Econometrica* 41 (3): 455–65.
- Dreze, Jean. 2011. "The Cash Mantra." *Indian Express*, May 11. <https://indianexpress.com/article/opinion/columns/the-cash-mantra/>.
- Dreze, Jean, and Reetika Khera. 2015. "Understanding Leakages in the Public Distribution System." *Economic and Political Weekly* 50 (7): 39–42.
- D'Souza, Anna, and Sharad Tandon. 2019. "Intrahousehold Nutritional Inequities in Rural Bangladesh." *Economic Development and Cultural Change* 67 (3): 625–57.
- Egger, Dennis, Johannes Haushofer, Edward Miguel, Paul Niehaus, and Michael W. Walker. 2019. "General Equilibrium Effects of Cash Transfers: Experimental Evidence from Kenya." NBER Working Paper 26600.
- Eli, Shari, and Nicholas Li. 2020. "Caloric Intake and Energy Expenditures in India." *World Bank Economic Review* 35 (4): 1057–75.
- Fackler, Paul L., and Barry Kent Goodwin. 2001. "Spatial Price Analysis." In *Handbook of Agricultural Economics*. Vol. 1, edited by Bruce L. Gardner and Gordon C. Rausser, 971–1024. Amsterdam: Elsevier.
- Fafchamps, Marcel. 1992. "Cash Crop Production, Food Price Volatility, and Rural Market Integration in the Third World." *American Journal of Agricultural Economics* 74 (1): 90–99.
- FCI. 2018. "Procurement Data for Current Marketing Season." FCI. <http://fci.gov.in/procurements.php?view=87> (accessed April 19, 2018).
- Fetzer, Thiemo. 2020. "Can Workfare Programs Moderate Conflict? Evidence from India." *Journal of the European Economic Association* 18 (6): 3337–75.
- Filmer, Deon, Jed Friedman, Eeshani Kandpal, and Junko Onishi. 2023. "Cash Transfers, Food Prices, and Nutrition Impacts on Ineligible Children." *Review of Economics and Statistics* 105 (2): 327–43.
- Finkelstain, Israel, and James A. Chalfant. 1991. "Marketed Surplus under Risk: Do Peasants Agree with Sandmo?" *American Journal of Agricultural Economics* 73 (3): 557–67.
- Finkelstain, Israel, and James A. Chalfant. 1997. "Commodity Price Stabilization in a Peasant Economy." *American Journal of Agricultural Economics* 79 (4): 1208–17.
- Finkelstein, Amy, Nathaniel Hendren, and Erzo F.P. Luttmer. 2019. "The Value of Medicaid: Interpreting Results from the Oregon Health Insurance Experiment." *Journal of Political Economy* 127 (6): 2836–74.
- Gadenne, Lucie. 2020. "Can Rationing Increase Welfare? Theory and an Application to India's Ration Shop System." *American Economic Journal: Economic Policy* 12 (4): 144–77.
- Gadenne, Lucie, Samuel Norris, Monica Singhal, and Sandip Sukhtankar. 2024. *Data and Code for: "In-Kind Transfers as Insurance."* Nashville, TN: American Economic Association; distributed by Inter-university Consortium for Political and Social Research, Ann Arbor, MI. <https://doi.org/10.3886/E1985621>.
- Gentilini, Ugo. 2016. *The Other Side of the Coin: The Comparative Evidence of Cash and in-Kind Transfers in Humanitarian Situations*. Washington, DC: World Bank.
- Ghatak, Maitreesh, and Karthik Muralidharan. 2020. "An Inclusive Growth Dividend: Reframing the Role of Income Transfers in India's Anti-Poverty Strategy." *India Policy Forum*.
- Ghatak, Maitreesh, Chinmaya Kumar, and Sandip Mitra. 2016. "Cash versus Kind. Understanding the Preferences of the Bicycle Programme Beneficiaries in Bihar." *Economic and Political Weekly* 51 (11): 51–60.
- Gibson, John, and Scott Rozelle. 2005. "Prices and Unit Values in Poverty Measurement and Tax Reform Analysis." *World Bank Economic Review* 19 (1): 69–97.
- Gollin, Douglas, and Richard Rogerson. 2014. "Productivity, Transport Costs and Subsistence Agriculture." *Journal of Development Economics* 107: 38–48.

- Gopalan, C., B. V. Rama Sastri, and S. C. Balasubramanian. 1989. *Nutrition Value of Indian Foods*. Hyderabad, India: National Institute of Nutrition, Indian Council of Medical Research.
- Government of India Planning Commission. 2005. *Performance Evaluation of Targeted Public Distribution System (TPDS)*. New Delhi: Government of India Planning Commission.
- Hanoch, Giora. 1977. "Risk Aversion and Consumer Preferences." *Econometrica* 45 (2): 413–26.
- Hari, Siddharth. 2016. "Markets, Subsidies, and Corruption: Evidence from India." Unpublished.
- Hastings, Justine, and Jesse M. Shapiro. 2018. "How are SNAP Benefits Spent? Evidence from a Retail Panel." *American Economic Review* 108 (12): 3493–3540.
- Haushofer, Johannes, and Jeremy Shapiro. 2017. "The Short-Term Impact of Unconditional Cash Transfers to the Poor: Experimental Evidence from Kenya." *Quarterly Journal of Economics* 132 (4): 2057–60.
- Hidrobo, Melissa, John Hoddinott, Amber Peterman, Amy Margolies, and Vanessa Moreira. 2014. "Cash, Food, or Vouchers? Evidence from a Randomized Experiment in Northern Ecuador." *Journal of Development Economics* 107: 144–56.
- Hirvonen, Kalle, and John Hoddinott. 2020. "Beneficiary Views on Cash and In-Kind Payments: Evidence from Ethiopia's Productive Safety Net Programme." *World Bank Economic Review* 35 (2): 398–413.
- Honorati, M., U. Gentilini, and R. G. Yemtsov. 2015. *The State of Social Safety Nets 2015*. Washington, DC: World Bank.
- Imbert, Clement, and John Papp. 2015. "Labor Market Effects of Social Programs: Evidence from India's Employment Guarantee." *American Economic Journal: Applied Economics* 7 (2): 233–63.
- Jensen, Robert T., and Nolan H. Miller. 2008. "The Impact of Food Price Increases on Caloric Intake in China." *Agricultural Economics* 39: 465–76.
- Jiménez-Hernández, Diego, and Enrique Seira. 2021. "Should the Government Sell You Goods? Evidence from the Milk Market in Mexico." Unpublished.
- Kaul, Tara. 2018. "Household Responses to Food Subsidies: Evidence from India." *Economic Development and Cultural Change* 67 (1): 95–129.
- Khera, Reetika. 2011. "Revival of the Public Distribution System: Evidence and Explanations." *Economic and Political Weekly* 46 (44-45): 36–50.
- Khera, Reetika. 2014. "Cash vs. In-Kind Transfers: Indian Data Meets Theory." *Food Policy* 46: 116–28.
- Kleibergen, Frank, and Richard Paap. 2006. "Generalized Reduced Rank Tests Using the Singular Value Decomposition." *Journal of Econometrics* 133 (1): 97–126.
- Kochar, Anjini. 2005. "Can Targeted Food Programs Improve Nutrition? An Empirical Analysis of India's Public Distribution System." *Economic Development and Cultural Change* 54 (1): 203–35.
- Kotwal, Ashok, Milind Murugkar, and Bharat Ramaswami. 2011. "PDS forever?" *Economic and Political Weekly* 46 (21): 72–76.
- Leroy, Jef L., Paola Gadsden, Sonia Rodriguez-Ramirez, and Teresa González De Cossío. 2010. "Cash and In-Kind Transfers in Poor Rural Communities in Mexico Increase Household Fruit, Vegetable, and Micronutrient Consumption but Also Lead to Excess Energy Consumption." *Journal of Nutrition* 140 (3): 612–17.
- Lieber, Ethan M. J., and Lee M. Lockwood. 2019. "Targeting with In-Kind Transfers: Evidence from Medicaid Home Care." *American Economic Review* 109 (4): 1461–85.
- Li, Nicholas. 2021. "In-Kind Transfers, Marketization Costs and Household Specialization: Evidence from Indian Farmers." Unpublished.
- Margolies, Amy, and John Hoddinott. 2015. "Costing Alternative Transfer Modalities." *Journal of Development Effectiveness* 7 (1): 1–16.
- Massell, Benton F. 1969. "Price Stabilization and Welfare." *Quarterly Journal of Economics* 83 (2): 284–98.
- McIntosh, Craig, and Andrew Zeitlin. 2021. "Cash versus Kind: Benchmarking a Child Nutrition Program against Unconditional Cash Transfers in Rwanda." Unpublished.
- Moffitt, Robert. 1989. "Estimating the Value of an In-Kind Transfer: The Case of Food Stamps." *Econometrica* 57 (2): 385–409.
- Montiel Olea, José Luis, and Carolin Pflueger. 2013. "A Robust Test for Weak Instruments." *Journal of Business and Economic Statistics* 31 (3): 358–69.
- Muralidharan, Karthik, Paul Niehaus, and Sandip Sukhtankar. 2017a. "Direct Benefits Transfer in Food: Results from One Year of Process Monitoring in Union Territories." Unpublished.
- Muralidharan, Karthik, Paul Niehaus, and Sandip Sukhtankar. 2017b. "General Equilibrium Effects of (Improving) Public Employment Programs: Experimental Evidence from India." NBER Working Paper 23838.

- Muralidharan, Karthik, Paul Niehaus, and Sandip Sukhtankar.** 2023. "Identity Verification Standards in Welfare Programs: Experimental Evidence from India." *Review of Economics and Statistics* 1–46.
- National Sample Survey Office of India.** 2005. Household Consumer Expenditure, NSS 59th Round [Dataset]. Government of India.
- National Sample Survey Office of India.** 2006. Household Consumer Expenditure, NSS 60th Round [Dataset]. Government of India.
- National Sample Survey Office of India.** 2007. Household Consumer Expenditure, NSS 61st Round [Dataset]. Government of India.
- National Sample Survey Office of India.** 2008. Household Consumer Expenditure, NSS 62nd Round [Dataset]. Government of India.
- National Sample Survey Office of India.** 2009. Household Consumer Expenditure, NSS 63rd Round [Dataset]. Government of India.
- National Sample Survey Office of India.** 2010. Household Consumer Expenditure, NSS 64th Round [Dataset]. Government of India.
- National Sample Survey Office of India.** 2012. Household Consumer Expenditure, NSS 66th Round [Dataset]. Government of India.
- National Sample Survey Office of India.** 2013. Rural Price Collection Survey, 2001–2011 [Dataset]. Government of India.
- National Sample Survey Office of India.** 2014. Household Consumer Expenditure, NSS 68th Round [Dataset]. Government of India.
- Newbery, David M.** 1989. "The Theory of Food Price Stabilisation." *Economic Journal* 99 (398): 1065–82.
- Newbery, David M., and Joseph E. Stiglitz.** 1981. *The Theory of Commodity Price Stabilization: A Study in the Economics of Risk*. Oxford: Oxford University Press.
- Nichols, Albert L., and Richard J. Zeckhauser.** 1982. "Targeting Transfers Through Restrictions on Recipients." *American Economic Review* 72 (2): 372–77.
- Niehaus, Paul, Antonia Atanassova, Marianne Bertrand, and Sendhil Mullainathan.** 2013. "Targeting with Agents." *American Economic Journal: Economic Policy* 5 (1): 206–38.
- NSSO.** 2001. *Nutritional Intake in India (Report No. 471)*. Delhi: National Sample Survey Organisation.
- Rao, B. S. N., and B. Sivakumar.** 2010. "Nutrient Requirements and Recommended Dietary Allowances for Indians." National Institute of Nutrition, Indian Council of Medical Research Report.
- Rao, M. Govinda.** 2019. "Redesigning the Fiscal Transfer System in India." *Economic and Political Weekly* 54 (31): 52–60.
- Rashid, Shahidur.** 2009. *Food Price Stabilization Policies in a Globalizing World*. Ithaca, NY: Cornell University Press.
- Rosenzweig, Mark R., and Christopher Udry.** 2014. "Rainfall Forecasts, Weather, and Wages Over the Agricultural Production Cycle." *American Economic Review* 104 (5): 278–83.
- Rosenzweig, Mark R., and Kenneth I. Wolpin.** 1993. "Credit Market Constraints, Consumption Smoothing, and the Accumulation of Durable Production Assets in Low-Income Countries: Investments in Bullocks in India." *Journal of Political Economy* 101 (2): 223–44.
- Rosenzweig, Mark R., and Oded Stark.** 1989. "Consumption Smoothing, Migration, and Marriage: Evidence from Rural India." *Journal of Political Economy* 97 (4): 905–26.
- Roy, Devesh, Ruchira Boss, and Mamata Pradhan.** 2020. *How India's Food-Based Safety Net is Responding to the COVID-19 Lockdown*. Washington, DC: International Food Policy Research Institute.
- Sabates-Wheeler, Rachel, and Stephen Devereux.** 2010. "Cash Transfers and High Food Prices: Explaining Outcomes on Ethiopia's Productive Safety Net Programme." *Food Policy* 35 (4): 274–85.
- Saha, Atanu, and Janice Stroud.** 1994. "A Household Model of On-Farm Storage Under Price Risk." *American Journal of Agricultural Economics* 76 (3): 522–34.
- Shapiro, Jeremy.** 2019. "The Impact of Recipient Choice on Aid Effectiveness." *World Development* 116 (C): 137–49.
- Shrinivas, Aditya, Kathy Baylis, and Ben Crost.** 2019. "Labor Market Effects of Food Transfers: Evidence from India's Public Distribution System." Unpublished.
- Shrinivas, Aditya, Kathy Baylis, Ben Crost, and Prabhu Pingali.** 2018. "Do Staple Food Subsidies Improve Nutrition? Evidence from India's Public Distribution System." Unpublished.
- Siu, Jade, Olivier Sterck, and Cory Rodgers.** 2021. "The Freedom to Choose: Theory and Quasi-Experimental Evidence on Cash Transfer Restrictions." Unpublished.

- Sotelo, Sebastian.** 2020. "Domestic Trade Frictions and Agriculture." *Journal of Political Economy* 128 (7): 2690–2738.
- Subramanian, Shankar, and Angus Deaton.** 1996. "The Demand for Food and Calories." *Journal of Political Economy* 104 (1): 133–62.
- Sukhtankar, Sandip.** 2017. "India's National Rural Employment Guarantee Scheme: What Do We Really Know About the World's Largest Workfare Program?" Unpublished.
- Tarozzi, Alessandro.** 2005. "The Indian Public Distribution System as provider of food security: Evidence from child nutrition in Andhra Pradesh." *European Economic Review* 49 (5): 1305–30.
- Turnovsky, Stephen J., Haim Shalit, and Andrew Schmitz.** 1980. "Consumer's Surplus, Price Instability, and Consumer Welfare." *Econometrica* 48 (1): 135–52.
- Victora, Cesar G., Linda Adair, Caroline Fall, Pedro C. Hallal, Reynaldo Martorell, Linda Richter, Harshpal Singh Sachdev, and Maternal and Child Undernutrition Study Group.** 2008. "Maternal and Child Undernutrition: Consequences for Adult Health and Human Capital." *Lancet* 371 (9609): 340–57.
- Waugh, Frederick V.** 1944. "Does the Consumer Benefit from Price Instability?" *Quarterly Journal of Economics* 58 (4): 602–14.
- Weaver, Jeff, Sandip Sukhtankar, Paul Niehaus, and Karthik Muralidharan.** 2023. "Cash Transfers for Child Development: Experimental Evidence from India." Unpublished.
- World Bank.** 2014. *The State of Social Safety Nets 2014*. Washington, DC: World Bank.